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Deep learning for financial forecasting: A review of recent trends☆

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ABSTRACT

Deep Learning (DL) has revolutionized financial forecasting, yet most reviews remain purely descriptive and lack actionable insight. This paper presents a comprehensive review of 187 Scopus-indexed studies (2020–2024) on DL applications for financial forecasting. We examine key DL architectures—Deep Multilayer Perceptrons, Recurrent Neural Network variants (Long Short-Term Memory (LSTM), Bidirectional LSTM, Gated Recurrent Units, and Reservoir Computing), Convolutional Neural Networks, Temporal Convolutional Networks, and Autoencoders. The studies are organized by forecasting task—stock, index, forex, commodity, bond, cryptocurrency, and volatility—and by model type (standalone vs. hybrid), preprocessing strategies, multi-modal feature integration (technical, fundamental, and sentiment signals), and novel methodological contributions.

Our findings confirm the dominance of RNN-based models, particularly LSTMs, while hybrid architectures combining convolutional and recurrent components (i.e., CNN-LSTM) are increasingly used to capture complex spatial–temporal dependencies. Stock and index forecasting remain prevalent, though cryptocurrency prediction is emerging due to continuous trading and high volatility. Integrating diverse signals (technical, fundamental, and sentiment) improves robustness but remains inconsistently applied.

To move beyond brand-name-focused literature, we propose a design-principles taxonomy encompassing sequential-memory frameworks, spatial–temporal hybrids, attention-augmented ensembles, preprocessing-augmented pipelines (e.g., EMD, VMD, wavelets), chaos/quantum-inspired hybrids, and application-oriented architectures supporting algorithmic trading, robo-advisors, and portfolio optimization. This taxonomy highlights design rationale, under-explored intersections, and practical guidance for model selection.

Critical gaps persist, including robustness under extreme market conditions, lack of standardized evaluation protocols, finance-specific interpretability, and computational constraints for real-time deployment. We outline a forward-looking research agenda emphasizing resilience, interpretability, scalability, lightweight deployment, modular hybrid models, behavioral insights, meta-learning, causal reasoning, emerging paradigms, such as chaos-inspired, and quantum-enhanced models, and real-time and continual learning. By framing DL in financial forecasting as a design-centered challenge rather than a black-box prediction task, this review aims to guide the development of adaptive, transparent, and high-performing financial systems.

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1. Introduction

Financial markets generate massive streams of complex, non-stationary data, creating both challenges and opportunities for predictive modeling. Accurately forecasting financial time series is critical for investment strategies, risk management, and policy decisions. Traditional econometric and statistical models often struggle to capture the nonlinear relationships, high volatility, and intricate temporal patterns inherent in financial data. In recent years, Deep Learning (DL) techniques have emerged as powerful alternatives, demonstrating the potential to model complex dependencies, adapt to evolving market dynamics, and improve predictive performance across diverse financial applications.

The fundamental motivation is to answer the following:

- Which DL models are preferred (and more successful) in different applications?
- What challenges do DL models face when applied to financial time series data?
- What is the future direction for DL research in financial forecasting?

Despite the growing adoption of DL in finance, existing surveys have several limitations. Most reviews focus on studies published prior to 2020, concentrate on specific asset classes or model families, or primarily provide descriptive overviews without systematic evaluation. As a result, there remains a need for a comprehensive, up-to-date synthesis that maps recent advancements, highlights prevailing trends, and identifies methodological challenges across financial forecasting domains.

This paper provides a systematic review of 187 Scopus-indexed studies on DL applications in financial time series forecasting, published between 2020 and 2024. The goal is to offer a comprehensive and holistic overview of recent advancements in DL-based financial forecasting. To the best of our knowledge, this study represents one of the first systematic surveys dedicated exclusively to literature from this period. The primary objectives of this review are the following:

- **Comprehensive coverage of financial domains:** We analyze and categorize 187 Scopus-indexed studies spanning stocks, indices, forex, commodities, bonds, cryptocurrencies, and volatility forecasting, providing a detailed overview of DL adoption in each domain.
- **Systematic taxonomy of models and methodologies:** In each financial forecasting domain, DL models are organized into standalone and hybrid architectures. We further examine feature integration (technical indicators, sentiment analysis, and alternative data), preprocessing techniques (dimensionality reduction, decomposition, denoising), optimization strategies, and novel methodological contributions.
- **Cross-study quantitative synthesis:** We tabulate representative studies by model type, dataset, feature set, performance metrics, and development environment, enabling clear comparisons and highlighting patterns in data usage, feature engineering, and methodological choices. Cross-study statistics summarize the findings within each forecasting domain.
- **Identification of trends, gaps, and actionable insights:** We identify prevailing trends, persistent challenges, methodological gaps, and emerging opportunities in financial DL research. Building on a design-principles taxonomy, this analysis shifts the focus from model labels to architectural rationale, providing actionable recommendations for developing robust, interpretable, and efficient context-aware DL models. The review also outlines clear directions for future research to advance the field.

Our analysis reveals several key insights. Stock and index forecasting remain dominant, while cryptocurrency prediction is an emerging domain. LSTM-based architectures continue to lead in performance, but hybrid models combining convolutional and recurrent components (i.e., CNN-LSTM) are increasingly prevalent. Feature integration, combining technical, fundamental, and sentiment data, enhances predictive performance but is inconsistently applied. Python is the primary language for implementing these DL models, due to its extensive libraries and broad adoption in financial AI research. Methodological challenges persist, including data scarcity and quality, model optimization, computational constraints for real-time deployment, robustness under extreme market conditions, and limited interpretability. Additional gaps include the lack of standardized evaluation protocols, insufficient stress-testing under market shocks, and finance-specific interpretability.

By integrating model architectures, application domains, and performance outcomes within a unified framework, this review provides a comprehensive, design-centered perspective on DL in financial forecasting. It offers actionable insights and a forward-looking research agenda emphasizing resilience, interpretability, scalable and lightweight deployment, real-time and continual learning, behavioral insights, meta-learning, causal reasoning, and emerging paradigms such as chaos- and quantum-inspired models. This review also serves as a structured reference for future studies and practical applications, bridging the gap between methodological development and real-world financial prediction needs.

2. Research design

To systematically collect relevant studies on DL models in financial forecasting, a structured search is conducted in Scopus database. The search was performed within the fields of *Article Title, Abstract, and Keywords* to ensure the retrieval of publications explicitly addressing the intersection of DL and financial forecasting.

The query included the term "*deep learning*" in conjunction with financial and predictive terms ("*financ**", "*forecast**", "*predict**") to capture studies within the broader domain of financial forecasting. To focus on DL architectures commonly used in time series forecasting, we included a set of specific model keywords: "*ANN*", "**MLP*", "*RNN*", "*LSTM*", "*RC*", "*ESN*", "*CNN*", "*Autoencoders*", *"GRU"*, and *"GAN"*. Furthermore, the query was restricted to studies applied to financial instruments, incorporating terms such as *"stock"*, *"index"*, *"forex"*, *"commodity"*, *"bond"*, and *"cryptocurrency"*.

The search was limited to publications from *2020 to 2024*. Additionally, to maintain relevance to the study's focus, only articles classified under the subject areas of *Computer Science, Engineering, Mathematics, Economics, Econometrics, and Finance* were considered. The final dataset was filtered to include only *English language journal articles*, excluding conference papers, reviews, and other document types.

A total of 211 research articles were initially collected. After excluding two review papers and filtering out studies that did not align with the scope of financial forecasting, 187 papers were selected, summarized, and categorized based on the application areas, including stock price forecasting, index forecasting, forex forecasting, commodity forecasting, bond forecasting, cryptocurrency forecasting, and volatility forecasting. The exclusions were due to factors such as restricted access, a focus on non-forecasting financial tasks (e.g., asset allocation, pricing, credit risk assessment, and corporate financial performance), non-financial forecasting tasks (e.g., NLP tasks, fraud detection, material demand forecasting, and agricultural exports prediction), or the use of non-relevant data types (e.g., accounting data, loan data, systemic risk indicators, sales demand forecast data, and material information). Although survey papers were not included in this analysis, two review publications, Sezer et al. (2020) and Ma'arif et al. (2024), were identified during the paper collection process and are discussed in the following section on existing surveys.

It is important to note that our study's inclusion and exclusion criteria introduce certain limitations. By restricting the dataset to peer-reviewed journal articles in English and excluding conference papers, gray literature, and preprints, some relevant findings may not have been captured. Similarly, the focus on specific model keywords and financial instruments may have omitted studies using alternative DL architectures or applied to less common asset classes. Readers should interpret the findings with awareness of these selection boundaries.

3. Existing surveys

Computational intelligence has long been applied in finance, with DL models emerging as particularly powerful tools, often surpassing traditional ML approaches. To date, a large body of surveys has attempted to consolidate progress in this area. Below, we discuss the most influential contributions, highlighting both their insights and their limitations.

One of the earliest comprehensive reviews bridging ML and DL, is by Rundo et al. (2019), who span statistical models, ML techniques, DL architectures, and hybrids, while touching on emerging areas like reinforcement learning, sentiment analysis, and quantum finance. Their contribution lies in cataloging the most commonly used datasets in financial forecasting, namely BSE, S&P 500, DAX, NASDAQ, SSE, and in flagging the lack of standardized benchmarks and consistent evaluation metrics, as a central obstacle to reproducibility and fair comparison. However, their analysis is largely descriptive, lacking algorithmic formulations with technical depth and little discussion of key challenges such as overfitting or interpretability. A more focused historical mapping is offered by Durairaj and Mohan (2019), who review 34 DL-based hybrid models between 1999 and 2019. They note the dominance of MLP-based hybrids compared to the fewer LSTM- or CNN-based architectures, highlight the use of evolutionary algorithms for network optimization, and note that most hybrids outperform standalone models. Yet, while their taxonomy is useful as a historical and methodological guide with chronological mapping of hybrid approaches, the review falls short in critically evaluating practical performance.

Among the most important contributions to the field are those of Ozbayoglu et al. (2020) and Sezer et al. (2020). Ozbayoglu et al. (2020) map DL architectures to applications ranging across algorithmic trading, risk assessment, portfolio management, fraud detection, derivatives markets. They emphasize the growing adoption of DL, especially in algorithmic trading and risk assessment, while highlighting trends like CNN–LSTM hybrids, and NLP ensembles for financial sentiment analysis, deep reinforcement learning, and cryptocurrencies. While they point to future directions such as cryptocurrencies, robo-advisory, sentiment analysis, high-frequency trading, and the potential of specific DL paradigms (e.g., 2D-CNNs, graph-CNNs, GANs, and transfer learning), their review overlooks methodological limitations and regulatory implications. Sezer et al. (2020), deliver a systematic literature review of 140 studies on DL methods for financial time series forecasting (2005–2019). They detail architectures, datasets, features, forecasting horizons, performance metrics, and development frameworks across asset types. The review finds that stock, trend, and index forecasting dominate DL research, with LSTMs leading due to their temporal modeling strengths, CNNs and DMLPs used mainly for classification, often requiring careful preprocessing or feature engineering, and a rising interest in deep reinforcement learning and cryptocurrency forecasting. However, they also caution against publication bias favoring DL, as newer DL studies may receive more attention. Future directions in hybrid DL–NLP, sentiment analysis, and adaptive reinforcement learning approaches are highlighted. These two works set the foundation for the current review, but both remain limited to pre-2020 studies.

Addressing the reproducibility concerns raised by earlier works, Buczyński et al. (2023) undertake a rare large-scale replication of 15 DL architectures across diverse assets and market periods comparing them against classical benchmarks such as naive forecasts, ARIMA, and exponential smoothing. Their findings are sobering: naive forecasts often rival DL methods in the long run, while only multi-scale recurrent convolutional neural network and ModAugNet architectures show short-term advantages. Accuracy varies by asset class, with currency pairs and indices easier to predict than volatile stocks and cryptocurrencies. The study highlights the fragility of many DL gains, the need for frequent retraining and proper tuning in non-stationary markets, and full methodological transparency, while underscoring ongoing reproducibility challenges. This study represents a landmark contribution, shifting the focus from descriptive cataloging toward rigorous evaluation.

Several subsequent surveys zoom in on specific model families or methodological issues. Thakkar and Chaudhari (2021) conduct a controlled experimental comparison of nine DNN on a unified dataset (S&P 500,2000–2017) using Independent Component

Analysis for feature extraction to reduce reliance on large sets of technical indicators. They find DQNs particularly effective for non-linear dynamics, while LSTM/GRU remain strong baselines. They stress the potential of hybrids and metaheuristic optimization, though their tuning remains exploratory. The need for standardized benchmarking to improve reproducibility and comparability is emphasized. Shah et al. (2022) concentrate on ARIMA, CNN, LSTM, and CNN–LSTM hybrids, showing that CNN–LSTM models are particularly effective, and sentiment integration adds a valuable behavioral dimension, enhancing short-term trends. They note key limitations: ARIMA's linearity assumptions, and outlier sensitivity, LSTM's vanishing-gradient issues and long runtimes, CNN's heavy data and computational demands, and hybrid models inheriting these constraints. Hyperparameter tuning and large historical datasets are also critical. Ma'arif et al. (2024) focus on LSTM hybrids for stock price prediction (2018–2024), covering optimization algorithms (e.g., GA-LSTM, LSTM-ARO), statistical/econometric techniques, preprocessing (e.g., EMD/EEMD, Wavelet), and feature extraction. GA-LSTM and CNN-LSTM are the most common, with cross-study RMSE comparisons, across diverse datasets and research contexts, identifying GA-LSTM, AGA-LSTM, and ARIMA-LSTM as top performers.

Broader reviews continue to appear, offering updated panoramas. Chen et al. (2023) reviews 85 works from 2015–2023 (identified via Web of Science), covering standalone models (CNN, LSTM, ensemble CNN) and hybrids (CNN-LSTM, LSTM-AM, CNN-LSTM-AM). LSTMs are acknowledged as the de-facto "should-have" element in DL frameworks, while hybrids—particularly CNN-LSTM and CNN-LSTM-AM—generally outperform standalone models. Practical aspects such as preprocessing, feature engineering, transfer learning, ensemble learning, validation, and explainable AI (e.g., LRP, SHAP) are noted, but the discussion is mostly generic. Key challenges include EMH-related delayed predictions, limited accessibility for finance domain experts, insufficient integration of domain knowledge, lack of standardized datasets, and difficulties with real-time or high-frequency prediction. Vuong et al. (2024) review 73 Scopus-indexed stock price forecasting studies, cataloging ARIMA, ML, DL, and hybrid methods, and report a steady rise in publications from 2014–2023, with a notable surge in DL and hybrid studies from 2020 to 2023. DL and hybrid models, particularly LSTMs, CNN–LSTMs, transformers, and GANs, excel at capturing non-linear, heterogeneous data, but face data requirements, architectural complexity, and overfitting challenges. The review overlooks alternative data sources, advanced techniques, and excludes non-journal literature, thereby limiting comprehensiveness. Nevertheless, it identifies future directions, such as alternative data integration (e.g., sentiment, news, real-time indicators), AI-driven architectures (such as graph CNN, reinforcement learning, and meta-learning), real-time analysis, and quantum computing. While informative as a descriptive survey, the study's scope and methodological constraints restrict its ability to provide a practice-oriented evaluation. Mienye et al. (2024) provide perhaps the broadest overview of DL models (e.g., RNNs, LSTMs, CNNs, Transformers, GANs, deep RL) and their applications across finance, including trading, risk management, fraud detection, forecasting, and portfolio optimization. The study highlights emerging trends such as explainable AI, transfer learning, federated learning, quantum computing, and ethical AI, while noting challenges like data quality, overfitting, computational complexity, bias, and regulatory concerns. Though lacking quantitative evaluations and empirical validation, the work's strengths lie in its broad coverage, connecting DL models to specific finance tasks, clear explanations of sophisticated concepts, and awareness of emerging trends and ethical considerations.

Certain reviews target niche but increasingly important aspects. Sonkavde et al. (2023) highlight ensemble methods, particularly combinations such as Random Forest, XGBoost, and LSTM, as the most promising, offering superior predictive accuracy. The study underscores the critical role of hyperparameter tuning, proposes integration of fundamental, technical, and sentiment analysis for improved forecasting, and suggests future directions such as trend analysis, and computer vision for pattern identification and candlestick chart pattern analysis. However, the evaluation is limited to two Indian stocks and primarily price-based features, restricting generalizability and overlooking richer auxiliary data sources. Li and Bastos (2020) systematically examine DL for technical analysis, showing LSTM dominance (73.5%), often in hybrid use with CNNs or sentiment analysis, but also critiquing the limited attention to profitability assessment (35.3%), risk management (rarely addressed), and trading design adopting simple buy-sell-hold logic, without adaptive/AI-based approaches, or real-market implementations. While thorough in cataloging DL implementations, the study offers limited critical analysis of performance differences, preprocessing, or hybrid impacts, but stresses the need for algotrading validation, cost-aware profitability, qualitative data, cross-market testing, and clearer guidance on technical indicators. Olorunnimbe and Viktor (2023) focus exclusively on studies with proper backtesting, finding that most research centers on trading strategy, price prediction (both often via DRL and LSTM), and portfolio management (with DRL particularly popular), while areas such as market simulation, stock selection, hedging, and risk management receive far less attention. Although several tasks have been framed as online learning, the connection remains largely unestablished, representing an opportunity to advance practice. Key challenges like limited data availability and quality, short-term prediction focus, trading costs neglect, and lack of specialized frameworks for online learning and non-stationary markets are acknowledged, which the authors suggest addressing through explainable models, public data availability, long-horizon perspectives, and dedicated financial DL frameworks to unify and mature the field. Blasco et al. (2024) through novel lens focus on uncertainty quantification (UQ). The authors contrast traditional point-prediction DL methods with UQ approaches that generate probabilistic forecasts, distinguishing aleatoric uncertainty, arising from inherent data noise, from epistemic uncertainty, stemming from the model and its specification. They note growing interest in UQ since 2010, peaking in 2019 with 389 studies, with most studies focusing on price accuracy and epistemic uncertainty. Gaps remain in addressing aleatoric uncertainty (or its combination with epistemic), integrating technical, fundamental, and sentiment data, lack of streaming data, multi-step, or trend-based predictions, and underutilization of markets such as Forex. Together, these specialized surveys expand the conversation from raw predictive accuracy to real-world considerations of backtesting, risk, and uncertainty.

Parallel to methodological reviews, a number of surveys emphasize applications. Hu et al. (2021) review 86 studies on stock and Forex forecasting, finding LSTM-based models, especially attention-LSTM and CNN-LSTM variants, dominate, while most rely on technical indicators and few incorporate sentiment, news, or expert recommendations. Wang and Yan (2021) analyze DL in

Table 1
Comparison of major surveys on deep learning and financial forecasting: scope, models, and reported contributions (Part I).

Survey	Scope/Period	Models covered	Main contributions
Rundo et al. (2019)	ML + DL in finance (pre-2019)	ARIMA, GARCH, SVM, RF, CNN, RNN, LSTM, hybrids	Broad review of statistical, ML, and DL models; mapping of common datasets; highlights need for benchmark datasets and evaluation metrics.
Durairaj and Mohan (2019)	34 hybrid DL models (1999–2019)	MLP-, LSTM-, CNN-based hybrids	Historical overview of MLP-, LSTM-, and CNN-based hybrid DL models; suggests combining DL with ML techniques, fuzzy logic, chaos theory, and evolutionary optimization.
Ozbayoglu et al. (2020)	First “deep learning for financial applications" survey	CNN, RNN/LSTM, DMLP, RBM, DBN, autoencoders	Overview of DL architectures; maps DL methods to financial tasks (trading, risk, portfolio, fraud, cryptocurrency); identifies trends in financial sentiment analysis, cryptocurrencies, DRL, and hybrid architectures.
Sezer et al. (2020)	First “financial time series forecasting” DL-only survey; 140 DL studies (2005–2019)	CNN, RNN, LSTM, RBM, DBN, autoencoders, DRL	Systematic review with structured taxonomy by asset type; stock, trend, and index forecasting dominate; LSTMs lead, CNNs/DMLPs for classification; rising DRL and cryptocurrency interest.

algorithmic trading, pointing to hybrid CNN–LSTM and DMLP-RL models for trading strategies, and DMLPs for regression and classification tasks with preprocessing. They note growing exploration of reinforcement learning and genetic algorithms for trading strategy development, while emphasizing the importance of parameter optimization, feature selection, and hybrid architectures, and warning on the security risks of widespread algorithmic trading given market fragility. Peng and Yan (2021) adopt a data-centric view of financial risk prediction, highlighting the challenges of heterogeneity (structured versus unstructured forms), imbalance (or information asymmetry between companies and stakeholders), and multi-source origins. They identify model interpretability, systemic risk propagation, and limitations of incomplete datasets as key areas for future research. Noting that some studies neglected data imbalance and omitted preprocessing techniques, like undersampling or oversampling, the authors stress the need for careful data handling in financial risk modeling. Finally, Qu et al. (2019) review ML/DL for bankruptcy prediction, framing it as a binary classification problem. Despite offering limited methodological insight and practical guidance, the authors emphasize the potential of heterogeneous data sources (e.g., news, reports) and identify model interpretability as an emerging concern.

Building directly on Sezer et al. (2020) and Ozbayoglu et al. (2020), the present review extends this line of inquiry to the more recent period of 2020–2024. Unlike prior surveys, it not only catalogues model architectures but also systematically maps research across asset classes, with tabulated summaries of datasets, feature sets, methodological approaches, and performance metrics. By integrating insights from various model architectures (e.g., standalone, hybrids, novel), preprocessing techniques, optimization strategies, and application domains, this review moves beyond descriptive reporting toward an analytical synthesis of the field. This approach reframes the literature through a design-principles-based taxonomy, highlighting prevailing trends, methodological challenges, and gaps, and actionable directions for future research in financial forecasting (see Tables 1–6

4. Deep Learning (DL)

In Fig. 1, we illustrate the position of Deep Learning, Machine Learning, and Artificial Intelligence.

DL technology originates from Artificial Neural Networks (ANNs), where the term “deep” refers to the number of hidden layers in the network. ANNs consist of interconnected neurons organized in layers. A simple ANN contains three layers: input, hidden, and output, as illustrated in Fig. 2. Information flows through the network via weighted connections. A bias term is added to the weighted inputs, and the sum is passed through an activation function, which determines the neurons responsible for feature extraction. The network output is compared to the target to compute the error, which is backpropagated to update the weights using an optimization algorithm. This process is repeated iteratively until convergence.

The mathematical formulation for a single-layer ANN is:

$$\mathbf{y} = f(\mathbf{W}\mathbf{x} + \mathbf{b}), \tag{1}$$

where:

- $\mathbf{y} \in \mathbb{R}^m$ is the output vector of predicted financial variables,
- $\mathbf{x} \in \mathbb{R}^n$ is the input feature vector (e.g., historical prices, technical indicators),
- $\mathbf{W} \in \mathbb{R}^{m\times n}$ is the weight matrix connecting inputs to outputs,
- $\mathbf{b} \in \mathbb{R}^m$ is the bias vector, and
- $f(\cdot)$ is an element-wise activation function (e.g., ReLU, sigmoid, tanh).

Table 2
Comparison of major surveys on deep learning and financial forecasting: scope, models, and reported contributions (Part II).

Survey	Scope/Period	Models covered	Main contributions
Buczyński et al. (2023)	Replication of 15 DL models	multi-scale recurrent CNN, ModAugNet, ARIMA, naïve, exponential smoothing	Large-scale reproduction across 29 assets and 5 market periods (2007–2022); unified evaluation framework; shows asset-dependent performance, and the need for frequent retraining and proper tuning of DL models.
Thakkar and Chaudhari (2021)	Stock price and trend forecasting (2017–2020)	CNN, RNN, LSTM, GRU, ESN, DNN, DQN, RBM, DBN	Panoramic review of DNNs; controlled experimental benchmark of nine models on S&P 500 (2000–2017); finds DQN effective and supports hybrids, and metaheuristic optimization.
Shah et al. (2022)	Stock prediction	ARIMA, CNN, LSTM, CNN–LSTM	Comparative review with flowcharts and tabulated summaries; highlights hybrid CNN–LSTM for combining feature extraction and temporal modeling; notes role of sentiment integration.
Ma'arif et al. (2024)	LSTM hybrids (2018–2024)	GA-LSTM, PSO-LSTM, APSO-LSTM, CNN-LSTM, ARIMA-LSTM, ARIMA-EGARCH-LSTM, LSTM-EMD, etc.	Yearly classification of LSTM hybrids; tabulated comparison of models, datasets, and metrics; identifies GA-LSTM, AGA-LSTM, and ARIMA-LSTM as top performers.
Chen et al. (2023)	85 DL studies (2015–2023)	CNN, LSTM, ensemble CNN, CNN–LSTM, LSTM-AM, CNN–LSTM–AM	Generalized framework for FTS prediction: architecture, training/prediction, hyperparameter tuning, and evaluation metrics; shows hybrids CNN–LSTM, CNN–LSTM–AM outperforming standalone.
Vuong et al. (2024)	73 Scopus-indexed stock price forecasting studies (2014–2023)	ARIMA, ML, DL, hybrids	Bibliometric and content analysis with tabulated summaries of methods, datasets, targets, features, metrics, and results; highlights LSTMs, CNN–LSTMs, transformers, and GANs as particularly effective.
Mienye et al. (2024)	DL in finance (broad)	feedforward network, RNN, LSTM, GRU, CNN, Transformer, GAN, DRL, DBN	Broad overview linking DL models to applications (trading, risk, fraud, portfolio); highlights trends in XAI, transfer learning, federated learning, quantum computing, and ethical AI.
Sonkavde et al. (2023)	Ensemble forecasting	linear regression, KNN, SVM, logistic regression, ARIMA, FB Prophet, LSTM, GRU, RF, XG-Boost	Comprehensive review of ML/DL methods; proposes ensemble (RF + XGBoost + LSTM) with superior performance on Indian stocks; emphasizes hyperparameter tuning.
Li and Bastos (2020)	DL + technical analysis (2017–2020)	34 DL models	Systematic review of DL for TA; analysis of predictor techniques, trading strategies, profitability metrics, and risk management; shows LSTM dominance and notes limited trading-strategy sophistication.
Olorunnimbe and Viktor (2023)	DL with backtesting (35 papers)	feedforward NN, RNN, CNN, autoencoder, DRL	Review restricted to studies with backtesting; classifies by task (trade strategy, price prediction, portfolio, risk); discusses use of market, fundamental, and alternative data; emphasizes domain-specific evaluation.
Blasco et al. (2024)	UQ[a] in DL for finance (69 studies, 2001–2022; Scopus, Google Scholar)	Probabilistic DL, UQ methods	PRISMA-based review; distinguishes epistemic vs aleatoric UQ; recommend richer uncertainty representations, enhanced feature spaces, and improved posterior approximation methods.
Hu et al. (2021)	DL for stock & Forex prediction (86 studies, 2015–2021)	CNN, RNN, LSTM, DNN, RL	Systematic review of stock + Forex prediction; highlights LSTM-based hybrids; notes limited sentiment integration.
Wang and Yan (2021)	DL in algorithmic trading (35 studies, 2016–2021)	CNN, LSTM, DMLP	Reviews DL models for prediction, and arbitrage; highlights importance of model selection and quality financial data.
Peng and Yan (2021)	DL for financial risk prediction	MLP, CNN, RNN, LSTM, BERT	Data-centric survey; emphasizes heterogeneity, imbalance, and multi-source data.
Qu et al. (2019)	Bankruptcy prediction (ML + DL)	Classical ML + DL	Short review tracing shift from statistical to ML/DL; notes role of heterogeneous data and model interpretability.

[a] Uncertainty Quantification.

Various kinds of DL models exist in the literature, including Deep Multilayer Perceptron (DMLP), Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), Gated Recurrent Unit (GRU), Reservoir Computing (RC),

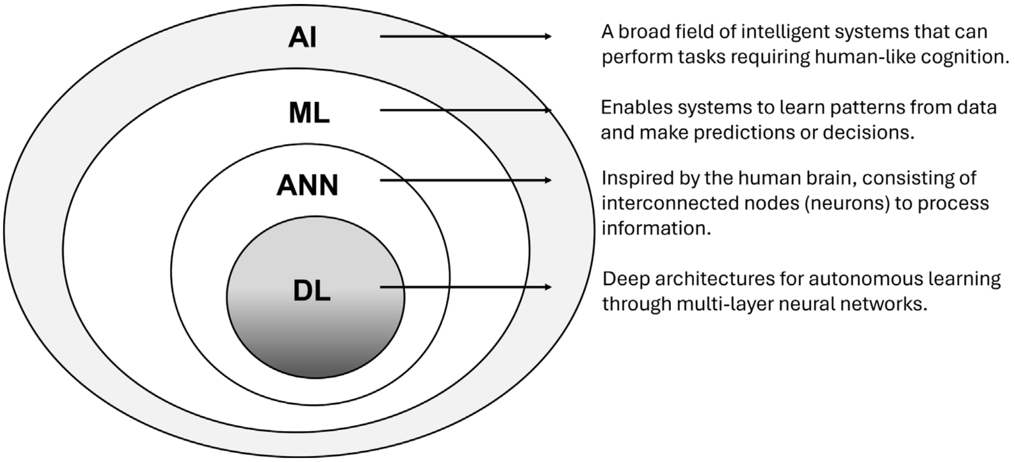

Fig. 1. A depiction of the relationship between deep learning (DL), machine learning (ML), and artificial intelligence (AI).

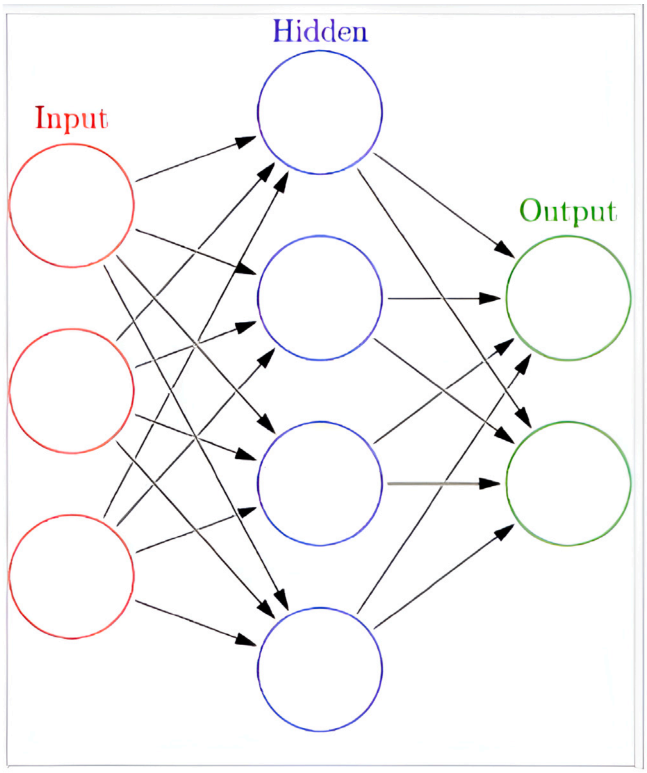

Fig. 2. Illustration of a simple ANN architecture, comprising the input, hidden, and output layers.

Convolutional Neural Network (CNN), Temporal Convolutional Network (TCN), AutoEncoders, other deep structures, and hybrid models. In what follows these are briefly discussed.

4.1. Deep Multi-Layer Perceptron (DMLP)

The Multi-Layer Perceptron (MLP) is a fully connected feedforward ANN consisting of an input layer, an output layer, and one or more hidden layers, which serve as the computational engines. MLP is regarded as the foundational architecture for all subsequent training algorithms within the ANN framework. As one of the earliest and simplest neural network models, MLPs stand as the backbone of DL (Rasekhschaffe & Jones, 2019). Frank Rosenblatt established the perceptron in 1957, a neural network able to

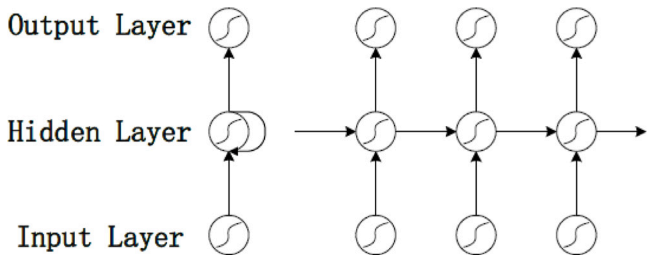

Fig. 3. Illustration of a standard RNN (left) and its unfolded version (right) (Zhuge et al., 2017).

classify images (Rosenblatt, 1958). Since then, advancements in computing power, increased data availability, and reduced capacity costs have facilitated the development of new techniques.

In DMLP, the deeper the network, the higher the complexity of the model and the higher the number of the hidden layers. In the hidden layers, each neuron has input (x), weight (w), bias (b) terms, and an activation function that produces the output of the network. The output of a single neuron in the network is depicted in Eq. (2).

The most widely used nonlinear activation functions are sigmoid (Eq. (3)) (Cybenko, 1989), hyperbolic tangent (Eq. (4)) (Kalman & Kwasny, 1992), Rectified Linear Unit (ReLU) (Eq. (5)) (Nair & Hinton, 2010), leaky-ReLU (Eq. (6)) (Maas et al., 2013), and softmax (Eq. (7)) (Ian et al., 2023). These activation functions introduce non-linearity into the network to learn complex mappings from data.

$$y_i = f\left(\sum_i w_i x_i + b_i\right) \tag{2}$$

where f is the activation function, chosen from:

$$\sigma(z) = \frac{1}{1+e^{-z}} \tag{3}$$

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \tag{4}$$

$$R(z) = \max(0, z) \tag{5}$$

$$R_{\text{leaky}}(z) = \alpha z \cdot 1(z < 0) + z \cdot 1(z \geq 0) \tag{6}$$

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \tag{7}$$

4.2. *Recurrent neural network (RNN)*

The RNN is a popular branch of ANN designed for processing sequential or time-series data. RNNs use outputs from previous time steps as inputs for the current step, enabling the model to rely on prior information in the sequence. The hidden layers of an RNN serve as memory, retaining information from earlier stages to forecast future steps. Fig. 3 illustrates the RNN architecture (left) and its unfolded representation (right).

At each time step t, the previous hidden output h_{t-1} and the current input x_t are used to compute the next hidden output h_t and the prediction $\hat{y}_t$, as given by the following equations:

$$h_t = \sigma\left(W_{hh} h_{t-1} + W_{hx} x_t\right) \tag{8}$$

$$\hat{y}_t = \text{softmax}\left(W_S h_t\right) \tag{9}$$

Here, W_{hx} represents the weight matrix for the input x_t, W_{hh} is the weight matrix for the previous hidden output h_{t-1}, and σ denotes the sigmoid activation function. Additionally, h_0 corresponds to the initial hidden output at time step $t = 0$. Eq. (9) represents the output probability distribution over the dataset at each time step t. Essentially, $\hat{y}_t$ is the next predicted element, given the sequential information up to that point and the most recent input element, x_t.

RNNs are well-suited for modeling sequential data and time series. However, they suffer from vanishing and exploding gradients, which hinder learning from long data sequences. In the case of vanishing gradients, the gradients shrink over time during backpropagation, leading to minimal weight updates. In contrast, exploding gradients occur when the gradients increase exponentially during backpropagation, causing the model's weights to grow uncontrollably. This results in an unstable model that is unable to learn during training from long input sequences (Pascanu, 2013). In the following, we discuss several popular variants of RNNs that address these challenges and perform well in many real-world applications.

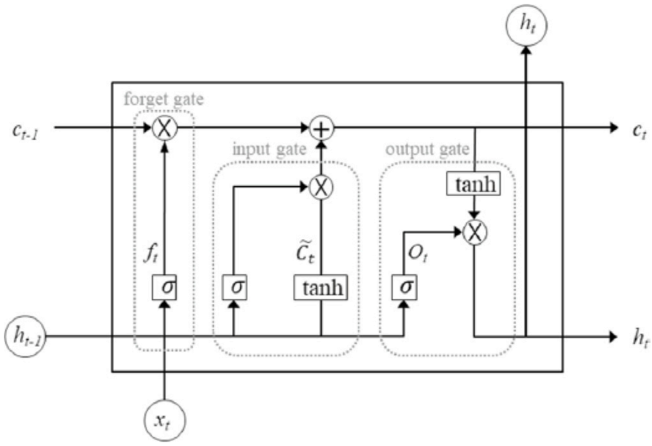

Fig. 4. Structure of an LSTM cell (Baek, 2024).

4.2.1. Long Short-Term Memory (LSTM)

The LSTM network, introduced by Hochreiter (1997), is a widely used variant of RNNs designed to deal with the vanishing gradient problem. It achieves this by employing gating mechanisms—the input, forget, and output gates—that regulate the flow of information within the network. The forget gate controls the retention or discarding of past cell state information, the input gate determines which new information should enter the cell state, and the output gate determines and controls the outputs. This design allows LSTMs to capture and retain long-term dependencies more effectively than standard RNNs. Fig. 4 illustrates the structure of an LSTM cell, where the upper horizontal line represents the cell state, facilitating information flow across time steps.

The gating mechanisms are analog and based on the sigmoid function, which operates within the range (0, 1), selectively filtering, adding, or removing information.

At each time step t, the forget gate regulates past information retention by applying a sigmoid function to the previous hidden state h_{t-1} and input x_t:

$$f_t = \sigma\left(W_{f,x}x_t + W_{f,h}h_{t-1} + b_f\right) \tag{10}$$

Next, the input gate updates the cell state. A candidate value $\tilde{c}_t$ is generated via a tanh activation, while i_t determines its contribution:

$$\tilde{c}_t = \tanh\left(W_{\tilde{c},x}x_t + W_{\tilde{c},h}h_{t-1} + \tilde{b}_c\right) \tag{11}$$

$$i_t = \sigma\left(W_{i,x}x_t + W_{i,h}h_{t-1} + b_i\right) \tag{12}$$

The updated cell state c_t is computed as:

$$c_t = f_t \circ c_{t-1} + i_t \circ \tilde{c}_t \tag{13}$$

Finally, the output gate regulates the hidden state update:

$$o_t = \sigma\left(W_{o,x}x_t + W_{o,h}h_{t-1} + b_o\right) \tag{14}$$

$$h_t = o_t \circ \tanh(c_t) \tag{15}$$

The weight matrices governing the forget, input, candidate, and output computations ($W_{f,x}, W_{f,h}, \tilde{W}_{c,x}, \tilde{W}_{c,h}, W_{i,x}, W_{i,h}, W_{o,x}, W_{o,h}$), along with their corresponding bias vectors ($b_f, \tilde{b}_c, b_i, b_o$), are optimized during training to minimize the loss function.

4.2.2. Bidirectional Long Short-Term Memory (BiLSTM)

BiLSTM, introduced by Graves and Schmidhuber (2005), processes sequential data in both forward and backward directions through two distinct hidden LSTM layers, as illustrated in Fig. 5. The forward layer processes the sequence in the same direction as the input, while the backward layer processes it in the reverse direction. BiLSTM connects both layers to the same output layer. The final output of the BiLSTM layer, y_t, is obtained by combining the forward and backward hidden states, as shown in Eq. (16), where the combination function σ can be concatenation, summation, averaging, or multiplication.

$$y_t = \sigma(\overrightarrow{h}_f, \overleftarrow{h}_b) \tag{16}$$

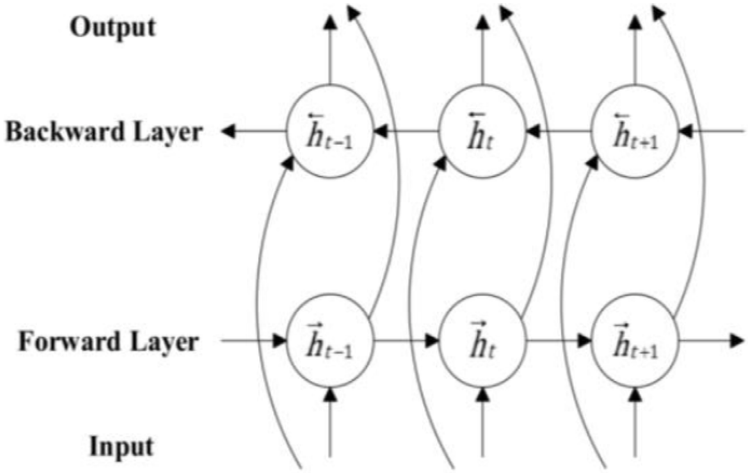

Fig. 5. Workflow of BiLSTM (Ferdiansyah et al., 2023).

4.2.3. Gated Recurrent Units (GRUs)

GRU, introduced by Cho et al. (2014), is another variant of RNN designed to mitigate the vanishing gradient problem. It utilizes gating mechanisms to control the flow of information between cells in the network, as illustrated in Fig. 6. GRU is a simplified variant of LSTM with fewer parameters. It uses two gates: the update gate and the reset gate, compared to LSTM's three gates (input, output, and forget gates) (Kumar et al., 2024). The update gate controls how much of the previous hidden state needs to be carried forward, enabling the GRU to capture long-range dependencies adaptively without discarding information from earlier parts of the sequence. The reset gate determines how much of the previous memory is combined with the new input in the current state calculation. The GRU transition equations are as follows:

$$z_t = \sigma\left(W_z \cdot [h_{t-1}, x_t]\right) \tag{17}$$

$$r_t = \sigma\left(W_r \cdot [h_{t-1}, x_t]\right) \tag{18}$$

$$\tilde{h}_t = \tanh\left(W \cdot [r_t \circ h_{t-1}, x_t]\right) \tag{19}$$

$$h_t = (1 - z_t) \circ h_{t-1} + z_t \circ \tilde{h}_t \tag{20}$$

The simpler structure of GRU leads to fewer matrix multiplications, enhancing its computational efficiency (Zhan & Kim, 2024). As a result, GRU enables faster training and tends to outperform LSTM when trained on limited datasets. In contrast, LSTM exhibits superior performance when dealing with larger datasets (Chen, 2023; Kaiser & Sutskever, 2015; Yin et al., 2017).

4.2.4. Reservoir Computing (RC)

ESN (Jaeger, 2001, 2007) is the virtual paradigm of RC derived from the theory of RNNs. The architecture of an ESN consists of an input layer, a reservoir, and a readout layer, as depicted in Fig. 7. The reservoir is a fixed, randomly initialized, and sparsely connected hidden layer of neurons, which maps input signals into a high-dimensional space. Generally treated as a "black box" (Cucchi et al., 2022), the reservoir serves as a high-dimensional dynamic system that transforms input signals into nonlinear internal representations, preserving temporal dependencies, the reservoir states. Using the notation in (Ballarin et al., 2022), the evolution of the reservoir state (x_t) for a given input sequence (z_t) is given by:

$$\begin{aligned} x_{t+1} &= lr \cdot x_t + \\ &\quad (1 - lr) \cdot f\left(\rho \frac{W_{res}}{\|\lambda_{\max}(W_{res})\|} x_t + \gamma \frac{W_{in}}{\|W_{in}\|} z_{t+1} + \omega \frac{\zeta}{\|\zeta\|}\right), \\ y_{t+1} &= W_{out}^{\top} x_{t+1} \end{aligned} \tag{21}$$

The key hyperparameters of the system include the leaking rate (lr), spectral radius (ρ), input scaling (γ), bias (ω), as well as the size of the reservoir (number of neurons) and the sparsity level. The weight matrices W_{in} and W_{res} remain fixed after initialization, while W_{out} is typically trained via linear regression (Dutoit et al., 2009; Lukoševičius & Jaeger, 2009):

$$W_{out} = (X^T X + \alpha I)^{-1} X^T Y \tag{22}$$

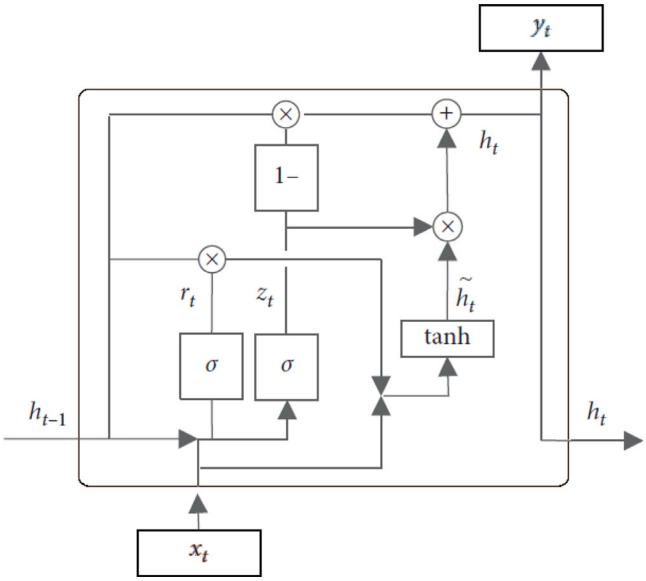

Fig. 6. Structure of the hidden layer node in GRU (Yang & Guo, 2021).

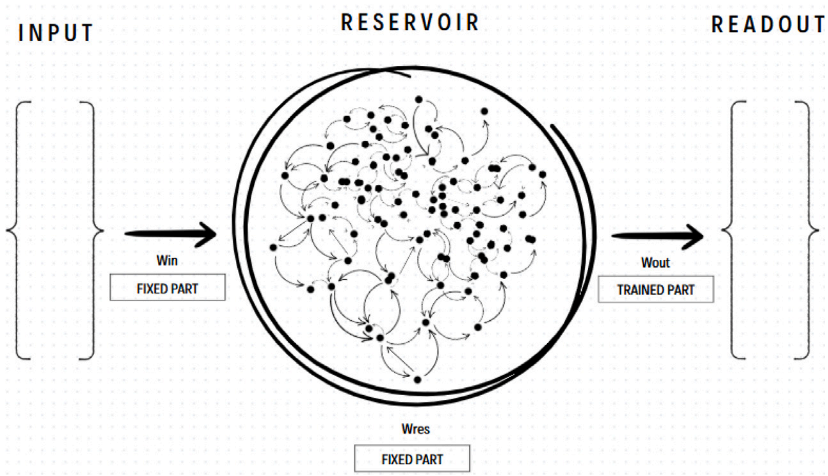

Fig. 7. ESN Architecture: The input layer processes external signals, which are then mapped into high-dimensional states by the reservoir. The output layer maps these states to the desired output, enabling predictions.

where X is the matrix of reservoir states collected over time, Y is the corresponding matrix of target outputs, and α is a regularization parameter that balances model complexity and generalization. ESNs offer a computationally efficient alternative for sequence modeling tasks, making them particularly relevant for financial forecasting applications.

A fundamental requirement for ESNs is that the reservoir satisfies the Echo State Property (ESP), ensuring that the influence of initial conditions vanishes over time. Although the condition $\rho(W_0) < 1$ is commonly cited to guarantee ESP, optimal system performance in practice depends on the design of the reservoir, hyperparameters, and overall system configuration. The ESP asserts that the reservoir dynamics maintain a stable "echo" state, preserving sensitivity to inputs without allowing their influence to grow exponentially over time. Achieving optimal performance requires careful hyperparameter tuning tailored to the task-at-hand. Since there are no universal design rules for ESNs, guidelines from Jaeger (2002), Lukoševičius (2012) offer practical insights, though empirical testing and task-specific adjustments remain essential.

In RC, reservoirs can be either virtual or physical. Independently and simultaneously with ESNs, Liquid State Machines (LSMs) (Maass et al., 2002) were developed as the physical paradigm of RC. LSMs employ spiking neurons, which more closely mimic the behavior of biological neurons compared to traditional artificial neurons found in feedforward or recurrent neural networks. These spiking neurons process information through discrete voltage spikes, triggered when the membrane potential surpasses a predefined threshold. Physical reservoirs, such as a bucket of water serving as a reservoir that makes computations on inputs given

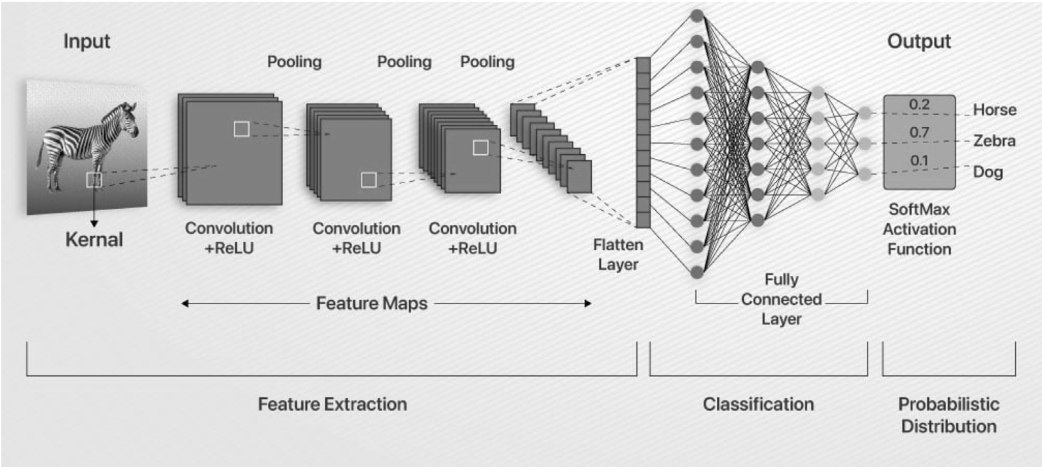

Fig. 8. CNN example.
Source: https://www.analytixlabs.co.in/blog/convolutional-neural-network/.

as waves on its surface (Fernando & Sojakka, 2003), are networks in the sense of continuous systems that operate in space and/or time.

Building upon LSMs, Liquid Neural Networks (LNNs) (Hasani et al., 2021) provide greater flexibility and adaptability through trainable neuron dynamics in both the reservoir and output layers, unlike ESNs and LSMs, where only the output layer is trained. LNNs are considered as Liquid Time Constant (LTC)-based architectures, where LTC is a critical parameter controlling the rate of information decay within the reservoir dynamics. LTC networks enhance the plasticity of DL models by enabling dynamic adaptation to new information, effectively balancing the retention of prior knowledge with the integration of new data. This adaptability prevents overfitting to old data and avoids forgetting previously learned tasks, akin to a liquid surface that adjusts to disturbances without losing its fluidity. Further analysis of physical RC is beyond the scope of this paper, but readers are suggested to follow Tanaka et al. (2019).

4.3. Convolutional Neural Networks (CNNs)

CNNs (LeCun et al., 1998) are discriminative DL architectures designed primarily for visual tasks such as image recognition, classification, and segmentation. The core components include the convolutional layer for feature extraction using filters or kernels, the pooling layer for dimensionality reduction through techniques like max or average pooling, summarizing clusters of neurons in one layer into a single neuron in the next layer (Bhatt et al., 2021), followed by a flattened layer, and the fully connected layer enabling the model to combine the extracted features for classification or prediction. Non-linearity is introduced via activation functions such as ReLU, sigmoid, or tanh (Ian et al., 2023). The output layer typically uses a logistic activation function, like softmax, to assign probability scores for classification (Bhatt et al., 2021; Kallurkar & Chandavarkar, 2024). Convolutional and pooling layers often form pairs, with multiple such pairs integrated into a CNN architecture, as shown in Fig. 8. CNNs are designed for processing 2D data, making them widely used in applications such as visual recognition, medical image analysis, and natural language processing.

The convolution operation extracts features from the input. For 1D data, the operation is defined as in Eq. (23), where t denotes time, $s(t)$ is the feature map, k is the kernel, x is the input, and a denote variable. For 2D image data, Eq. (24) describes the convolution, where I is the input image, K is the kernel, (m, n) represents image dimensions, and (i, j) denote variables. The neural network architecture follows in Eq. (25), where W_{ij} are the weights, x_j is the input, b_i is the bias, and z_i is the output of neuron i. The softmax function, as given in Eq. (26), is then applied to generate the final output, which represents the class probabilities for each y_i.

$$s(t) = \sum_{a=-\infty}^{\infty} x(a)k(t-a) \tag{23}$$

$$S(i,j) = \sum_{m}\sum_{n} I(m,n)K(i-m, j-n) \tag{24}$$

$$z_i = \sum_{j} W_{ij}x_j + b_i \tag{25}$$

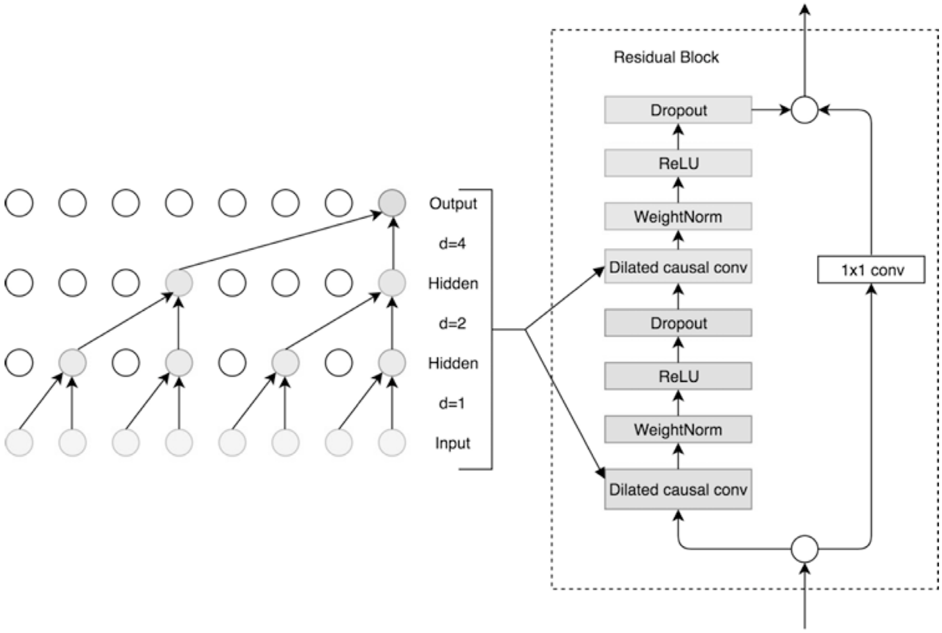

Fig. 9. TCN Architecture. The left part is the dilated causal convolution with a kernel size of 2 and dilation factors $d = [1, 2, 4]$. The right part is the residual block (Qin, 2019).

$$y_i = \text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \tag{26}$$

4.3.1. Temporal Convolutional Network (TCN)

Introduced by Lea et al. (2016), TCNs were further popularized and formalized by Bai et al. (2018a) for broader applications. Belonging to the family of CNNs, TCN is an extension designed to efficiently capture hidden temporal dependencies in sequential data. A TCN maps an input sequence x to an output of the same length using a 1D fully convolutional network architecture with zero-padding of size $(\text{kernel\_size} - 1)$ to maintain sequence length (Bai et al., 2018a). Causal convolutions ensure that each output at time t depends only on inputs from time t and earlier, preventing information leakage from future to the past (Bai et al., 2018a). To capture long-term dependencies efficiently, TCNs employ dilated convolutions (Van Den Oord et al., 2016), where dilation factor d controls the spacing between filter elements, increasing the receptive field exponentially. A dilated convolution F on an element s of the 1D sequence x is given by

$$F(s) = \sum_{i=0}^{k-1} f(i) \cdot x_{s-d \cdot i}, \tag{27}$$

where k is the filter size. Additionally, TCN utilizes residual blocks (He et al., 2016) to enhance stability in deeper architectures. Each residual block comprises two layers of dilated causal convolutions, weight normalization, ReLU activation, and spatial dropout. A 1×1 convolution operation is also incorporated in each residual block to ensure input–output size consistency. A dilated causal convolution with a kernel size of 2 and dilation factors of $[1, 2, 4]$, is illustrated in Fig. 9.

4.4. Autoencoder (AE)

An AE (Goodfellow, 2016) is a type of neural network designed for unsupervised learning, primarily employed in dimensionality reduction and feature extraction. It comprises two main components: an encoder, which maps the input into a lower-dimensional latent representation, and a decoder, which reconstructs the input from this compressed representation, as illustrated in Fig. 10. Given an input $x \in [0, 1]^d$, the encoder compresses it into a latent representation h, preserving essential features while discarding redundant information. This transformation is governed by a function $f(x)$, parameterized by a weight matrix W_1, a bias vector b_1, and a nonlinear activation function σ_1, typically a sigmoid or ReLU function. The decoder then reconstructs the input from the latent representation using an inverse function $g(h)$, which maps h back to the input space. This process involves another set of parameters: weight matrix W_2, bias vector b_2, and an activation function σ_2. The goal is to generate an output r that closely approximates the original input x. The model is trained by minimizing the reconstruction loss, typically measured using the MSE. Mathematically, the encoding and decoding processes are expressed as:

$$h = f(x) = \sigma_1(W_1 x + b_1) \tag{28}$$

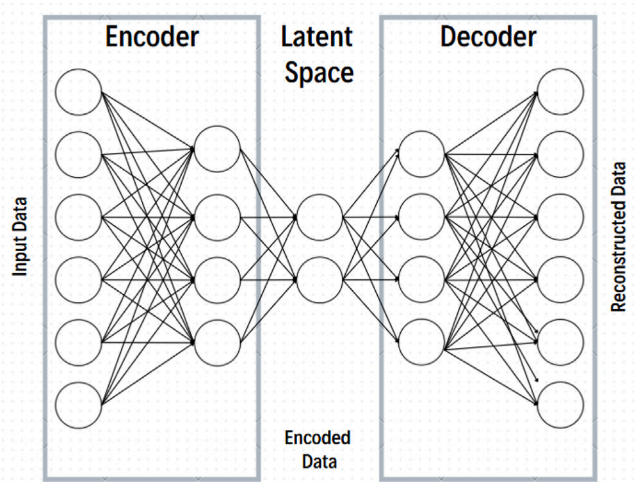

Fig. 10. Standard architecture of AE.

$$r = g(h) = \sigma_2(W_2 h + b_2) \tag{29}$$

$$L(x, r) = \|x - r\|^2 \tag{30}$$

AEs are widely applied in feature learning, anomaly detection, and data compression, leveraging their ability to learn compact and meaningful representations of input data.

While encoder–decoder models (Cho et al., 2014) share a similar structure with autoencoders, they differ in that they are designed for supervised tasks involving input–output transformations, mapping an input to a distinct output, such as in sequence generation or machine translation. In contrast, autoencoders are unsupervised models focused on reconstructing the input from a compressed latent representation, with the input and output being the same.

4.5. Other deep structures

Generative Adversarial Networks (GANs) (Goodfellow et al., 2014) are a class of neural networks designed for generative modeling, capable of producing realistic data samples by learning the underlying distribution of a given dataset. A GAN consists of two competing networks: a generator G, which generates synthetic data resembling real samples, and a discriminator D, which evaluates whether a given sample is real or generated (Liu et al., 2024a). These networks are trained in an adversarial process, where the generator aims to produce increasingly realistic data to deceive the discriminator, while the discriminator improves its ability to distinguish real from fake samples. The training process converges at an ideal point known as the Nash equilibrium, where the discriminator can no longer reliably differentiate between real and generated data. At this point, the generator has successfully learned the underlying distribution of the original data (Liu et al., 2023). GANs have been widely applied in image synthesis, data generation, and style transfer.

Transformers (Vaswani, 2017) are a class of DL models designed to handle sequential data efficiently by leveraging self-attention mechanisms. Unlike traditional recurrent architectures that process data sequentially, transformers establish direct connections between all elements in a sequence, enabling parallelization and improved long-term memory retention (Fischer et al., 2024). This architecture allows information to propagate across the entire sequence, improving performance in natural language processing and time series forecasting tasks. However, since all input elements are interconnected, the model relies on a self-attention mechanism to selectively focus on the most relevant information while filtering out noise, enabling context-aware learning. Several specialized variants have emerged for time series forecasting, including the Temporal Fusion Transformer (TFT) for multi-horizon and multivariate forecasting (Wu et al., 2024), and PatchTST (Nie et al., 2022) employing patching, where time series data is segmented into sub-sequences that serve as input tokens, and channel independence, wherein each channel corresponds to a single univariate time series sharing the same embedding and Transformer weights (Fan & Zhang, 2024).

Neural Basis Expansion Analysis for Time Series (N-BEATS) (Oreshkin et al., 2019) is a pure DL architecture designed for univariate time series forecasting, without relying on domain-specific components such as recurrent or convolutional structures. It outperformed the M4 competition-winning ES-RNN model (Makridakis et al., 2020) by 3%. N-BEATS is structured as a collection of stacks, each containing multiple blocks. Each block, serving as the basic processing unit, is a multi-layer fully connected network with ReLU activations, producing two outputs: a forecast for the target horizon and a backcast, which is the model's best estimate of its input given its functional limitations. The layer of blocks are combined together using a hierarchical doubly residual stacking topology. N-BEATS offers several advantages, including interpretability, ease of implementation across different domains without requiring modifications, and efficient training. Unlike many traditional time series models, it does not require specialized feature engineering or input scaling, making it a flexible and powerful approach to time series forecasting.

Trellis networks (Bai et al., 2018b) are temporal convolutional networks designed for sequence modeling, characterized by two key features: first, weights are shared not only by all time steps but also by all network layers, tying them into a regular trellis pattern; second, the input at any given time step is fed into all layers of the network, not just the first layer. Bridging recurrent and convolutional architectures, they leverage techniques from both domains. Recent implementations include SA-TrellisNet for news-driven stock index prediction with Sentiment Attention (Liu et al., 2024b) and GAN-TrellisNet, a multi-factor model integrating GANs, TrellisNet, and CNNs to capture temporal dependencies and refine predictions (Liu et al., 2024a).

A **Deep Q-Network (DQN)** (Mnih et al., 2015) is a reinforcement learning algorithm that combines Q-learning with deep neural networks to estimate and optimize action-value functions (Q-functions), enabling decision-making in high-dimensional state spaces. The core concepts of DQN include Q-learning, experience replay, and target networks. DQN uses Q-learning to approximate the optimal action-value function, mapping states to expected future rewards. Experience replay involves storing past experiences in a memory buffer and randomly sampling them during training to break temporal dependencies and stabilize learning. The target network is a separate network with the same architecture as the main network, periodically updated by copying the weights from the main network, which helps to stabilize learning. This approach has shown significant success in complex environments, such as Atari games.

4.6. Hybrids

Hybrid models, integrating DL, ML, and statistical methods, have become integral to financial forecasting. By combining the strengths of different modeling approaches, hybrid models can mitigate the limitations of individual methods, improving predictive accuracy, robustness, and adaptability. This improved performance, however, often comes at the expense of increased computational requirements and model complexity, necessitating careful model design, tuning, and evaluation. Examples include CNN-LSTM models (Abdullah & Salah, 2023; Aldhyani & Alzahrani, 2022; Hao & Gao, 2020; Mohamed Rida et al., 2024) which combine convolutional layers that extract short-term temporal features with recurrent layers that model long-term dependencies, LSTM-GRU hybrids for short-term cryptocurrency prediction (Patel et al., 2020), and optimization-enhanced DL frameworks like PSO-optimized LSTM/GRU (Chauhan et al., 2023) and GA-LSTM (Chen & Zhou, 2020). The applications, contributions, and limitations of the various hybrid approaches are discussed in the following sections across multiple financial forecasting domains.

5. Hyperparameters tuning

Hyperparameters are critical in defining the architecture of DL models and have a significant impact on their predictive performance. Optimizing these parameters is essential for obtaining optimal results in financial forecasting tasks (Abdullah et al., 2024). Practitioners should choose the optimization method that best aligns with the complexity of the search space, the available computational resources, and the specific characteristics of the model. Commonly employed hyperparameter tuning techniques are outlined below.

Manual tuning (which relies heavily on the practitioner's expertise), Random Search (which explores the hyperparameter space randomly), Grid Search (which exhaustively evaluates all combinations of hyperparameters), and Bayesian Optimization (a probabilistic, model-based approach) remain widely used for hyperparameter selection.

More advanced optimization techniques inspired by nature have gained popularity due to their ability to efficiently explore complex search spaces. Examples include evolutionary and swarm intelligence methods. The Artificial Rabbit Optimization (ARO) algorithm, for instance, draws inspiration from rabbit survival strategies, while Particle Swarm Optimization (PSO) mimics the social behavior of birds flocking and fish schooling. Adaptive PSO further refines this approach by dynamically adjusting parameters for faster convergence. Similarly, Artificial Bee Colony (ABC) optimization simulates the foraging behavior of honeybees, the Flower Pollination Algorithm (FPA) is inspired by the pollination behavior of flowers, Genetic Algorithms (GA) replicate natural selection processes, and Cuckoo Search leverages the brood parasitism strategy of cuckoo birds.

6. Financial time series forecasting

The application of DL models in finance is vast and diverse. In the previous section, we introduced the most prominent DL models. This section provides a review of DL models used in financial forecasting applications. We categorize the implementation areas and present them in separate subsections to offer a comprehensive overview. The aim is to summarize the most recent papers, specifically those published between 2020 and 2024, within the field of DL for financial forecasting.

In each subsection corresponding to a specific asset type (e.g., stock price forecasting, index prediction, forex price prediction, commodity price prediction, bond price forecasting, cryptocurrency price forecasting, and volatility forecasting), we further classify the studies based on their focus. We distinguish among standalone and hybrid models, novel approaches, technical and sentiment analysis-based models, and those employing preprocessing techniques like dimensionality reduction, decomposition, and optimization.

Moreover, in each subsection, we have tabulated the representative models, datasets, and features of the relevant studies to provide as much detailed information as possible. See the Online Appendix (available at: Download here). Each table includes the following fields to provide the implementation details of the papers within the group: Article (Art.) and Dataset refer to the study and the data used, respectively. Period indicates the time frame for training, validation, and testing. Feature Set lists the input features used in the study. Method specifies the DL models used. Performance Criteria provides the evaluation metrics, and Environment (Env.) includes the development framework, software, or tools employed. In some cases, certain columns may be left empty if the corresponding information was not available in the paper. The goal is to highlight the dominant models, features, and data types that stand out within each application area and to briefly discuss how each research article contributes to the existing body of literature.

Note: Some studies evaluate their models across multiple asset classes. To maintain the autonomy of each section and accurately reflect the trends within each domain, these studies are included in all relevant sections (e.g., stock forecasting, index forecasting, etc.). Consequently, overlap between sections is expected, and certain studies may be cited in multiple asset-specific subsections.

Note: Each subsection concludes with a summary highlighting key observations, cross-study statistics, and thematic insights from the reviewed literature. These summaries cover trends in model usage, feature selection, forecasting targets, geographic focus, data timeframes, and performance metrics. Summaries are omitted in cases where the number of reviewed studies is insufficient (typically ≤ 10) to support meaningful aggregation or generalization.

6.1. Stock price forecasting

Stock price forecasting has become a central application area for DL, driven by the growing demand for accurate, data-driven financial decision-making. The literature reflects a transition from monolithic models, such as ARIMA and other statistical approaches with fixed assumptions, to more flexible and task-specific designs, each tailored to asset characteristics and market regimes. This evolution includes both standalone and hybrid models, using RNNs, CNNs, attention mechanisms, and generative frameworks. There is also increased integration of technical indicators, sentiment analysis, dimensionality reduction, and optimization strategies to improve accuracy. This section reviews these advances, covering architectural innovations, feature integration, optimization, and applications in varied financial contexts, including pandemic resilience and specialized prediction tasks, organized according to the outlined subsections.

6.1.1. Standalone architectures

In what follows, recent literature is systematically reviewed and analyzed to highlight the main trends and developments in standalone DL models for stock forecasting.

Comparative performance of RNN variants. LSTM remains a dominant baseline model. Moradi et al. (2023) show that LSTM outperforms traditional ML models such as RF and SVR on Tehran Stock Exchange data, while Furizal et al. (2024) demonstrate improved accuracy with multivariate LSTM models. Despite its widespread use, recent studies report that GRU and ESN frequently outperform LSTM in accuracy and convergence speed. For instance, Sun et al. (2021) highlight ESN's efficiency over LSTM, and Lawi et al. (2022), Ismailova et al. (2024) find GRU consistently more effective. Jing and Pathmanathan (2023) further reveal GRU's strength in handling volatile assets, with LSTM performing better in less volatile ETFs. In contrast, models without gating mechanisms, such as VRNNs, and structurally different models like CNNs, often underperform due to limited ability to capture temporal dependencies.

Importantly, model suitability varies by forecasting horizon; LSTM excels in short-term forecasts compared to traditional technical stock market analysis models (e.g., SMA, EMA) but suffers from higher errors in long-term forecasts (Billah et al., 2024). Probabilistic models like HMMs can rival or even outperform LSTM and ensemble-based classifiers under certain conditions (Sher et al., 2023).

Novel model architectures. Several studies explore novel architectures and enhancements. Yang (2021) use LSTM and GRU to forecast financial indicators (ROTA, PTSR, PER) achieving low error rates with comparable accuracy. Md et al. (2023) propose a multi-layer sequential LSTM to mitigate the vanishing gradient problem, outperforming existing DL models. Gajjar et al. (2024) compare LTC-based architectures with RNN variants, including their bidirectional versions, finding that LNN and particularly BiLTC deliver superior forecasting performance, highlighting LTC as promising alternatives. Similarly, Zhan and Kim (2024) improve LSTM-based prediction by integrating time-window and convolutional filters to better capture both short- and long-term dependencies in financial data.

Integrating technical and sentiment analysis. Recent works also emphasize feature augmentation, combining technical indicators (e.g., RSI, EMA, ATR) (Dhokane & Agarwal, 2024; Patil et al., 2023; Sivadasan et al., 2024) and sentiment analysis (tweets, reports) (Song et al., 2020; Srivastava et al., 2023) with LSTM. These studies report consistent improvements. Notably, advanced variants such as the Improved Recurrent Rider LSTM (IRRLSTM) (Patil et al., 2023), integrating feature selection and data augmentation techniques, has shown superior results.

Other use cases. Beyond price prediction, studies have expanded DL use cases. Abuein et al. (2024) predict optimal trading times using SVR and LSTM. Velmurugan and Indhumathy (2020) treating future support and resistance values as a time-series problem, apply the Fibonacci sequence with LSTM, achieving high accuracy and valuable trading insights. GAN-based models (e.g., TimeGAN) address the differences of time-constrained stock markets versus continuous cryptocurrency markets (Santoro & Grilli, 2022), revealing higher predictive power in cryptocurrencies. Meanwhile, models like CNN-BI (Sezer & Ozbayoglu, 2019) and 2D-CNN (Tatane et al., 2024) leverage image-based representations of stock data (i.e., time series, bar charts, technical indicators) to enhance trading signal classification and improve trading decisions. These diverse approaches illustrate the expanding versatility of DL techniques in capturing complex market behaviors beyond traditional price forecasting.

6.1.2. Hybrid architectures

This part provides a systematic review and analysis of recent literature, focusing on the key trends and advancements in hybrid DL models for stock forecasting.

Fusion of CNNs, RNNs, and autoencoders. Several studies respond to the limitations of standalone models by proposing hybrid DL architectures that enhance forecasting accuracy. Models like CNN-LSTM (A & N, 2024; Abdullah & Salah, 2023; Aldhyani & Alzahrani, 2022; Mohamed Rida et al., 2024) and CNN-LSTM-GRU (Naufal, 2023) effectively capture multi-scale dependencies. Others, such as KM-CAE-DLSTM (Guan et al., 2024) and NR-LSTM (Song et al., 2024), incorporate clustering or trend-residual separation to improve learning. Additionally, signal denoising techniques, such as wavelet transforms, have proven to be effective preprocessing steps for enhancing the performance of LSTM-based forecasting models (Jarrah, 2024; Ma et al., 2024).

Novel model architectures. Novel studies on automated trading increasingly integrate DL with domain-specific enhancements to balance predictive accuracy and interpretability. Hybrid models like the VAR-LSTM by Sakib and Mustajab (2024) capture both linear and nonlinear dependencies that identify buy and sell signals by analyzing market regimes defined by moving averages correlation. High-frequency forecasting via ODE-LSTM (Li et al., 2024) improves dynamic feature learning and model interpretability, achieving strong returns in the Chinese A-share market for small-cap stocks. Ansari et al. (2022) introduce a decision support system using DRL with a GRU-based forecasting network, integrated with past stock price trends to form an enriched observation state for the DRL agent, enabling informed trading decisions, offers fast and strong predictive performance, with stable profitability.

Li et al. (2023) present a clustering-enhanced DL framework that improves stock return forecasting by grouping time-series with similar patterns using LWDTW, demonstrating that such clustering significantly boosts model performance, with LSTM outperforming RNN, and GRU, and other similarity measures. For high-frequency forecasting, Roy Choudhury et al. (2020) propose LSTM-VAE that denoise input data and predict short-term price deviations from recent averages, consistently outperforming traditional time-series models, while Zaznov et al. (2024) fuse convolution layers and GRU in TFF-CL-GRU for price direction prediction, using market microstructure data, boost predictive and trading performance at a high computational cost. Rooted in chaos theory, the novel PSR-Transformer (Phien & Platos, 2023) combines Phase Space Reconstruction (PSR) with Transformer encoders, uncovering hidden patterns through time-delay embedding and effectively capturing long-range dependencies. Addressing signal noise, Akşehir and Kılıç (2024) present ICE2DE-MDL, which combines two-level decomposition, entropy-based IMF classification as either high-frequency or noise-free, and model-specific forecasting (LSTM, GRU, SVR), demonstrating robust performance but increasing model complexity. These works reflect a broader trend: combining DL with signal processing, clustering, or regime-awareness can enhance performance, but often entails trade-offs between interpretability, scalability, and computation.

Several works introduce novel multi-modal hybrid architectures that combine temporal, spatial, and attention-based mechanisms. For instance, BiCuDNNLSTM-1dCNN (Kanwal et al., 2022) merges bidirectional LSTM with 1D-CNN to extract temporal information and abrupt changes, while AE-ACG (Li et al., 2023) integrates CNN-GRU blocks within an autoencoder for hierarchical attention-guided decoding, enhancing information utilization. Similarly, Rath et al. (2024) combine stacked Bi-LSTM, attention modules, and wavelet-denoised CNNs, further optimized using an equilibrium strategy. These models highlight an increasing trend toward modular, specialized architectures tailored for complex financial signals. Beyond classical DL, quantum-inspired methods like QGAF (Xu, Wang, et al., 2024) demonstrate promising improvements in stock trend prediction by transforming time series into high-quality grayscale images, showcasing the growing potential of quantum computing in financial forecasting. Collectively, these studies underscore both the innovation and fragmentation in model design, calling for comparative benchmarks and generalization-focused evaluations.

Enhancing prediction via dimensionality reduction. To address feature redundancy and improve forecasting accuracy, several studies incorporate dimensionality reduction techniques. Ding et al. (2023) apply GPCA with Grey Relation Analysis to extract key macroeconomic indicators and reduce input dimensionality, revealing profitability disparities across Chinese petroleum firms. Similarly, Mansoor et al. (2025) combine Mutual Information for feature ranking with PCA to streamline inputs for LSTM, improving prediction efficiency.

Optimization-integrated models. Several studies emphasize hyperparameter tuning as a performance bottleneck and apply bio-inspired optimization methods. Gülmez (2023) use the ARO algorithm to optimize LSTM for DJIA stocks. Other studies deploy PSO, FPA, GA, and Tree-structured Parzen Estimator methods (Kumar & Haider, 2021; Sivapurapu, 2020), with optimized hybrids improving intraday stock prediction, and generalize better across both short- and long-term forecasting horizons.

Novel hybrid optimization models combine metaheuristics and DL for improved stock forecasting. For instance, Pai et al. (2023) introduce GSO-LSTM, which outperforms traditional tuning methods like grid and random search. Salama (2025) presents the SHOAGAI-TSF framework that combines SHOA with Generative AI and CGAN for probabilistic prediction. Jayanth and Manimaran (2024) develop DES-DA-ED-Bi-GRU-BO, incorporating trend decomposition and Bayesian Optimization, while Liu and Long (2020) propose a hybrid system uniting EWT, dropout-LSTM, ORELM, and PSO, outperforming nine benchmark models.

Integrating technical and sentiment analysis. A growing body of work incorporates multi-modal inputs, combining price data with technical indicators and sentiment to capture richer market signals and improve forecasting robustness.

Uçkan (2024) propose a PCA-LSTM-CNN hybrid that enhances short-term prediction by identifying key technical indicators via PCA. Fan and Peng (2022) develop a DRL-based model combining DQN and LSTM to make optimal trading decisions using raw and technical features. To better capture market dynamics, Zhang et al. (2023) design a CNN-Bi-LSTM with attention and a modified loss function to enhance trading signals by adapting to the heavy-tailed distribution of stock price fluctuations. Chen and Zhou

(2020) introduce GA-LSTM, using market, technical, and financial factors. Addressing the limited data problem for newly listed stocks, AUDA-Net by Wei and Dai (2023) leverages GAN-based Adversarial Unsupervised Domain Adaptation for trend prediction. The model generates synthetic unlabeled data and uses adversarial domain adaptation to transfer knowledge from source to newly listed stocks, achieving accuracy comparable to supervised DL models despite limited labeled data. Liu et al. (2024a) propose GAN-TrellisNet, integrating GANs, CNNs, and TrellisNet architectures with an enriched alpha158+OCHLVC feature set, which enhances learning depth and outperforms benchmarks across global markets over a decade.

Beyond technical factors, sentiment analysis has become a crucial complementary input. Chandola et al. (2023), Ko and Chang (2021), Kumar et al. (2022) integrate sentiment extracted via BERT, Word2Vec, and Twitter polarity, respectively, with LSTM-based models. Eslamieh et al. (2023) go further by introducing User2Vec, a model that combines market data with Twitter-based sentiment and user credibility scores, improving accuracy, particularly for stocks with high social engagement. Choi et al. (2023) design a feature-interaction module to blend price and text features effectively, while Jin et al. (2020) integrate sentiment from bullish/bearish comments with EMD decomposition and attention-LSTM. Other works incorporate denoising techniques (Ilyas et al., 2022) and news sentiment via FinBERT (Chen et al., 2024) to further enhance multimodal fusion. These studies demonstrate that combining technical, and sentiment analysis with financial data consistently improves predictive performance.

Financial forecasting in pandemic contexts. The COVID-19 pandemic offered a natural experiment to test the resilience and adaptability of financial forecasting models. Notably, Lachaab and Omri (2023) find that KNN outperforms LSTM and ARIMA across pandemic phases, implying that non-parametric models may better handle sudden shifts in market behavior. Their conclusion subtly challenges the assumption that complex models like LSTM are always superior. Conversely, Nazareth and Reddy (2024) affirm LSTM's reliability for forecasting the sector-specific S&P BSE GREENEX Index, while emphasizing the importance of adjusting training window lengths to evolving market conditions, with longer windows during the pandemic and shorter ones post-pandemic to better capture shifting data dynamics. In our view, the key takeaway is that model effectiveness is context-dependent, and robustness under crisis conditions may favor simpler or dynamically tuned approaches.

Addressing forecasting challenges with specialized techniques. Several studies have brought forward significant observations and findings. For instance, the Three-Point Moving Gradient (TPMG) method by Shoong et al. (2022), an enhanced LSTM-based approach that predicts price gradients rather than absolute values, offering improved directional insight into market trends. AGNN (Wei et al., 2022) addresses market-style factor interference by using LSTM with attention and a cooperative network architecture, where subnetworks share a loss function based on stock rankings. The model outperforms ARIMA and LSTM, though it requires prior knowledge to guide effective interactions. N-BEATS (Zein et al., 2024) improves both performance and interpretability by incorporating past and future trends through backward-forward residual links and a deep stack of fully connected layers.

Fan and Zhang (2024) compare multiple DL architectures (LSTM, GRU, CNN-LSTM, PatchTST) for both forecasting and nowcasting, showing that LSTM performs best overall. They also find that EMD is more effective for forecasting than nowcasting, and models generally perform better in nowcasting. Wu et al. (2020) propose a novel parameterized Continuous Trend Labeling (CTL) method that improves return-based performance by refining how trends are labeled across models. For stock correlation forecasting, Luo et al. (2024) design the novel CLATT model, integrating CNN, BiLSTM, and attention, with Bayesian hyperparameter tuning. Finally, Wang, Lin, et al. (2024) introduce the Multi-Dimensional Spatio-Temporal Fusion Transformer (MDSTFT) architecture, combining CNN, LSTM-attention, and Transformer layers to model complex spatial and temporal relationships in multivariate financial time series.

While these models demonstrate superior forecasting capabilities in controlled experiments, their generalization across market regimes and their computational demands remain open challenges.

6.1.3. Summary

The reviewed literature reflects a maturing yet fragmented landscape of DL applications in stock price forecasting. While LSTM remains a dominant baseline, empirical results increasingly favor alternatives such as GRU, ESN, and novel architectures like BiLTC and IRRLSTM, especially when tailored to specific market conditions or enriched with technical and sentiment features. This diversity underscores the field's methodological innovation but also complicates cross-study comparability. The model effectiveness is highly context-dependent and is shaped by forecast horizon, data modalities, and asset volatility, highlighting the need for standardized benchmarks and rigorous ablation studies to disentangle architectural gains from feature engineering.

Recent literature on hybrid DL models for stock price prediction reveals a maturing convergence of architectures that integrate temporal, spatial, and attention-based components with preprocessing strategies such as denoising, dimensionality reduction, and clustering. These models aim to exploit the complementary strengths of their constituent modules, for instance, CNNs for extracting local patterns from raw price series, LSTM/GRUs for capturing long-term temporal dependencies, and attention mechanisms to selectively focus on the key information.

While many hybrid models (e.g., CNN-LSTM, LSTM-GRU, CNN-Transformer) achieve superior performance over traditional baselines, these benefits often come with trade-offs. Most notably, enhanced performance frequently incurs higher computational costs and greater sensitivity to hyperparameters. Moreover, interpretability tends to decline as models become more complex, particularly in architectures that fuse deep layers with multiple data sources (e.g., sentiment, financial indicators, and price signals), as it becomes harder to trace how input features influence the final predictions.

Although empirical results are promising, many studies focus on narrow asset classes or regions, limiting generalizability. Additionally, benchmarking remains inconsistent across works, with considerable variation in datasets, feature selection, and

Table 3
Cross-study overview of stock price forecasting: Models, assets, features, and timeframes.

Category	Top Entities/Statistics
Most Common Models	Standalone: LSTM (31), ML (13), GRU (9); Hybrids: LSTM-based (15), CNN-LSTM-based (8). Total applications: 117[a b].
Standalone vs. Hybrid Use	Standalone: ∼98%; Hybrid/ensemble models: ∼52%; Novel architectures: ∼21%.[c]
Frequent Feature Types	Nearly all studies (∼99%) use price-based features (OHLCV, Close, Adjusted Close); ∼23% incorporate technical indicators; ∼13% include sentiment (e.g., news, tweets). Notably, ∼13% (10 studies) rely solely on Close price data. Rare features include stock bar chart images, financial factors, and correlation data.
Most Common Assets Forecasted	US stocks (e.g., Apple, Tesla, Amazon), Chinese A-shares, Indian stocks (e.g., Axis Bank), Indonesian equities (e.g., PT Bank Central Asia).
Geographic Focus	USA (NASDAQ, NYSE, S&P500, etc.): ∼47% (35 studies); China (SSE, SZSE, CSI300): ∼15% (11 studies); India (NSE, BSE): ∼8% (6 studies); Indonesia: ∼3% (2 studies); Multi-region/global: ∼13% (10 studies).
Time Periods Covered	Most studies span 2000–2024, with a focus on 2010–2022. Long-term periods (>10 years) are used in ∼45% (34 studies), while ∼21% (16 studies) examine short periods (<5 years). Some datasets go back to the 1960s.
Sentiment-Integrated Models	10 studies combine market data with sentiment (from Twitter, news, forums). LSTM variants are most used (e.g., ABC-LSTM, Attention-LSTM), along with hybrid and embedding-based models (e.g., User2Vec).
Performance Metrics Used	The most commonly reported evaluation metrics include RMSE (∼56%, 42 studies), MAE (∼47%, 35), MAPE (∼39%, 29), and MSE (∼37%, 28). R^2 appears in ∼29% (22 studies). Classification-oriented metrics such as Accuracy (∼15%, 11), F1-Score (∼13%, 10), Recall (∼12%, 9), and Precision (∼11%, 8) are less frequent.

[a] Some studies use multiple models (e.g., baselines, ensembles), contributing to total model count > number of papers (75). This reflects broader comparative experimentation.
[b] Model groups are defined as follows: LSTM-based (e.g., LSTM-ARIMA, PSO-LSTM, VAR-LSTM); CNN-LSTM-based (e.g., CNN-LSTM, CNN–LSTM–Transformer, CNN-BiLSTM).
[c] Percentages represent the frequency of model appearances across the 75 studies. Since multiple models can be used within a single study, either as primary methods or for benchmarking, overlaps in counts are expected.

performance metrics. Less common metrics, some drawn from trading contexts, are used sporadically, showcasing the models' performance in real-life financial scenarios.

The rise of modular hybrid models, combining tailored preprocessing, architecture, and optimization, reflects growing flexibility in design. However, this also fragments the field, as varied setups hinder comparability. Unified evaluation protocols and broader cross-market testing are needed to assess generalizability and guide future model development.

Cross-study observations. Across the reviewed studies (n = 75), the most frequently employed models are standard (non-optimized) LSTM and LSTM-based hybrids (≈ 61% combined), followed by optimization-integrated architectures (≈ 11%), attention-enhanced models (≈ 9%), and clustering- or decomposition-based frameworks (≈ 7%). Despite the growing sophistication of hybrid designs, persistent challenges include computational complexity and sensitivity to hyperparameter settings.

6.2. Index forecasting

Rather than forecasting the price of individual stocks, many researchers focus on forecasting stock market indices, which tend to exhibit lower volatility due to their diversified composition, offering a better reflection of overall market momentum and economic conditions.

6.2.1. Standalone architectures

The following section presents a systematic review and analysis of recent literature, emphasizing key trends and advancements in standalone DL models for stock market index forecasting.

LSTM-based approaches. Recent research on standalone DL models for financial forecasting highlights the continued prominence of recurrent architectures, particularly LSTM networks, with effective performance across tasks and market conditions. For instance, Ta et al. (2020) demonstrate that an LSTM-based approach to portfolio construction yields higher returns and more reliable risk management compared to classical approaches such as Equal-Weighted allocation, Monte Carlo Simulation, and Mean-Variance Optimization. This suggests that LSTM can effectively balance predictive power and decision support in investment contexts. LSTM also exhibits superior predictive performance over alternative architectures, outperforming MLP in forecasting cross-sector fluctuations of the Tadawul index (Al-Nefaie & Aldhyani, 2022b), and surpassing both RNN and CNN models in Nifty50 index prediction (Fathali et al., 2022).

Nevertheless, architectural refinements on LSTM are increasingly reported. Zhan and Kim (2024), for example, improve LSTM's forecasting accuracy by employing the Tumbling Window technique with convolutional filters (Short-term Convolutional Filter), albeit at the cost of increased training time. In another direction, Tong and Yin (2022) take a step toward decision-aware forecasting by integrating LSTM with reinforcement learning, offering promising results for adaptive trading strategies that respond dynamically to market conditions. This aligns with broader literature emphasizing LSTM's robustness in index prediction tasks.

Beyond LSTM architectures. GRU, CNN, and TCN models have also demonstrated competitive performance. For instance, Yang and Guo (2021) demonstrate that GRU can outperform traditional statistical models such as ARMA, BVAR, and BP in forecasting inflation trends in China, highlighting its strength in capturing nonlinear temporal dynamics with fewer parameters than LSTM. CNN-based models have also evolved in response to common limitations; to mitigate information loss during image transformation, Kirisci and Cagcag Yolcu (2022) propose a modified CNN architecture specifically designed for financial time series, improving forecast accuracy for indices such as TAIEX and FTSE. Notably, Yilmaz and Yildiztepe (2024) find that TCNs outperform LSTM, GRU, ARIMA, and Random Walk models in stock return forecasting. Their success is largely attributed to the TCNs' causal, dilated convolution structure, which allows for efficient modeling of long-term dependencies without the vanishing gradient issues typically faced by recurrent networks.

Integrating technical analysis. Several studies employ technical indicators to enhance model performance. Manjunath et al. (2021) evaluate DL models for Nifty50 price prediction using two technical indicator datasets: TA1 (SMA, EMA, MACD, RSI, SMI, RVI, EMV, RN) and TA2 (SMA, EMA, MACD, RSI). Comparing three GRU variants with RNN and LSTM models, they find that GRU with TA1 performs best, emphasizing the great relevance of SMI and RN for index prediction. In contrast, Yan et al. (2021) investigate the impact of different input features, and conclude that basic trading indicators and the CSI300 Index contribute most significantly to prediction performance. They further observe that technical indicators may lead to information loss, while financial indicators provide limited value for short-term forecasting.

6.2.2. Hybrid models

The following section reviews recent literature on hybrid DL models for stock market index forecasting.

Fusions of LSTM and beyond. A growing body of literature explores hybrid models combining statistical and DL techniques. Donghwan et al. (2020) present P-LSTM, a pattern-driven model that combines BIRCH clustering with LSTM, enabling the training of separate LSTM models on similar data subsequences to enhance stability in heterogeneous financial data. Song et al. (2024) propose NR-LSTM, separating trend and residual components for improved bearish market performance. Other works (Li et al., 2020; Xiao & Su, 2022) follow a similar hybrid paradigm, coupling ARIMA with ML/DL models to capture linear–nonlinear patterns. Jia et al. (2024) propose ResNLS, integrating ResNet for feature extraction and LSTM for temporal modeling, while Anjum et al. (2023) propose a CNN-LSTM-RNN architecture to tackle noise, non-linearity, and volatility in financial data. Beyond LSTM-based models, ESNs have shown superior predictive accuracy and efficiency for non-stationary chaotic (Vosvrda macroeconomic system) and financial datasets (S&P 500 index) (Georgopoulos et al., 2023).

Dimensionality reduction for forecast enhancement. Mansoor et al. (2025) propose using Mutual Information for feature selection, PCA for dimensionality reduction while preserving significant data variations, and LSTM for improved stock market forecasting. Similarly, Wang et al. (2023) integrate PCA with an Improved GRU (IGRU) model, incorporating an Antioversaturation Conversion Module to enhance learning sensitivity and prevent oversaturation.

Novel model architectures. Recent innovations reveal a growing trend toward hybrid models that blend decomposition, optimization, and chaos theory. Notable innovations include the hybrid CNN-LSTM model by Hao and Gao (2020), which captures multi-scale features for trend forecasting; the Polylinear Regression with Augmented LSTM Neural Network (PLR-ALSTM-NN) by Ahmed et al. (2022), designed for multivariate time series forecasting with strong long-term predictive capabilities; and the LSTM-BN model by Fang et al. (2023), which enhances high-frequency movement forecasting by integrating Batch Normalization, an adaptive cross-entropy loss function, and a multi-path parallel structure with a voting mechanism.

Models like BiCuDNNLSTM-1dCNN (Kanwal et al., 2022) and Multi-Filters Neural Network (MFNN) (Long et al., 2019) effectively combine CNN and RNN elements to capture both spatial and temporal dependencies in financial data. Others, such as the ResNet18-attention model (Noh et al., 2023) and AE-ACG (Li et al., 2023), introduce advanced feature extraction and attention mechanisms to address multivariate complexity and enhance temporal representation. Notably, the use of DRL with GRU (Ansari et al., 2022) highlights a shift toward decision-oriented forecasting, prioritizing not just prediction but actionable outcomes for automated trading. The SF-Transformer (Mao et al., 2024) marks a promising step in integrating domain-specific knowledge, like Spot-Forward parity, into Transformer architectures, suggesting improved long-term forecasting accuracy and reduced uncertainty. Zhang et al. (2024) develop a 1D-CapsNet-LSTM that enhances long-horizon forecasts outperforming LSTM, RNN, and CNN-LSTM models.

Extending beyond LSTMs, advanced architectures like the Deep Convolutional GAN (DCGAN) (Staffini, 2022) combine a CNN-BiLSTM generator with a CNN discriminator to simulate trader learning. The generator produces initial price predictions, which are iteratively refined based on discriminator feedback comparing predictions to actual data. This model outperforms ARIMAX-SVR, RF, LSTM, and standard GANs in both single- and multi-step forecasting tasks. Wu et al. (2020) introduce Continuous Trend Labeling, improving classification accuracy and yielding higher net returns than traditional labeling and buy-and-hold strategies. Maratkhan et al. (2021) present three hybrid models, CNN with Technical Indicator Clustering (CNN-TIC), Evolutionary Optimized CNN-TIC

(EO-CNN-TIC), and Residual Network with Technical Analysis (ResNet-TA), which incorporate *take profit* and *stop loss* techniques for automatic trading, with EO-CNN-TIC further optimized using Cuckoo Search for better trading precision.

Novel Decomposition-Driven Architectures

Towards decomposition-based architectures, a variety of hybrid models, such as EMD/CEEMD-based CNN-LSTM (Rezaei et al., 2021), and Akima-EMD-LSTM (Ali et al., 2023), demonstrate the efficacy of decomposing input signals into intrinsic mode functions (IMFs) for denoising and dimensionality reduction prior to sequential modeling.

More advanced architectures, including CEEMDAN-TCN-GRU-CBAM (Li et al., 2024c) and GRU-CEEMDAN-Wavelet (Qi et al., 2023), enhance forecasting by applying fuzzy entropy for IMF reorganization and wavelet denoising to suppress noise in high-frequency components, thereby improving feature quality and model performance. In a broader study, Yañez et al. (2024) demonstrate that Transformer-based models, particularly when combined with CEEMDAN decomposition, consistently outperform LSTM-based, CNN-LSTM, and non-decomposed variants across global stock indices, with CEEMDAN-CNN-Transformer achieving top performance in 70% of cases. This highlights the combined benefit of frequency decomposition and advanced attention-based architectures. Meanwhile, leveraging entropy-based analysis, the Hierarchical Decomposition-based Forecasting Model (HDFM) (Li et al., 2024) integrates CEEMDAN for initial signal decomposition, K-means clustering of IMFs based on entropy levels, and VMD for further decomposition of high-frequency components, before employing GRU for final predictions. Similarly, Akşehir and Kılıç (2024) propose ICE2DE-MDL, leveraging a two-level decomposition strategy through ICEEMDAN and entropy-driven IMF classification, ensuring the retention of valuable high-frequency components while eliminating noise.

Novel Decomposition-Optimization-based Architectures

Several novel methods integrating decomposition, and optimization techniques have been proposed to enhance stock price forecasting. Liu and Long (2020) introduce the EWT-dpLSTM-PSO-ORELM model, combining EWT for pre-processing, dropout-enhanced LSTM (dpLSTM) for forecasting, PSO for hyperparameter tuning, and ORELM for error correction post-processing. Deng et al. (2022) propose MEMD-LSTM, leveraging MEMD to decompose multivariate time series into high-frequency, low-frequency, and residual signals, addressing issues such as mode mixing and scale misalignment inherent in conventional EMD methods, and LSTM optimized via orthogonal array tuning method. Wang, Yuan, et al. (2024) develop SSA-PSO-LSTM, integrating SSA for decomposing and reconstructing the stock price time series into independent components, PSO for LSTM optimization, and feature selection of inputs, including technical indicators, based on Pearson correlation coefficients. Yan et al. (2020) propose CEEMD-PCA-LSTM for one-step-ahead stock price prediction, integrating CEEMD for noise reduction and decomposition into IMFs, PCA for dimensionality reduction and feature extraction, and LSTM for sequential forecasting.

Novel Chaos-Driven Approaches

Through novel insights, chaos-based hybrid models have been developed to better capture the complex, noisy, and nonlinear dynamics of financial time series. Durairaj and Mohan (2021) introduce the Chaos-LSTM-PR, integrating Chaos Theory, LSTM, and PR to enhance prediction accuracy across foreign exchange, commodity prices, and stock indices. The approach begins by testing the FTS for chaotic behavior using the Lyapunov exponent. Chaos Theory is then applied to model the chaotic dynamics of the time series by reconstructing the phase space. For the creation of the phase space, it is necessary to determine the ideal lag from the time series, and the embedded dimension of the associated space. The Akaike Information Criterion (AIC) is used to determine the optimal lag from time series and Cao's method is used to achieve the optimal embedding dimension. The modeled time series serves as input for an LSTM model to generate initial forecasts. The sequence of errors derived from LSTM forecasts is PR appropriate for error predictions. Error forecasts and original model forecasts are applied to produce the final hybrid model forecasts. The proposed Chaos-LSTM-PR outperforms ARIMA, Prophet, LSTM, and Chaos-LSTM, in terms of MSE, D-statistic, and Theil's U. Building on this, Durairaj et al. (2023) further validate the Chaos-LSTM-PR model on the same datasets over an extended period, comparing it against a broader range of models, including ARIMA, Prophet, LSTM, CNN, CART, RF, Chaos-CART, Chaos-RF, and Chaos-LSTM, using metrics such as MSE, MAPE, Dstat, and Theil's U. The findings reaffirm Chaos-LSTM-PR's superiority, consistently outperforming all counterparts. Similarly, Durairaj and Mohan (2022) propose Chaos-CNN-PR, which leverages CNN for initial forecasts of chaos-checked financial time series, refined by PR. This model achieves superior performance over traditional models (ARIMA, Prophet, CART, RF, CNN) and chaos-augmented models (Chaos-CART, Chaos-RF, Chaos-CNN) in forecasting exchange rates, commodity prices, and stock market indices.

Signal decomposition and denoising architectures. Several studies leverage decomposition techniques to improve the modeling of complex dynamics in financial time series. Fan and Zhang (2024) utilize EMD to isolate underlying temporal patterns, finding that while EMD enhances forecasting by capturing long-term structures, it offers limited benefits for nowcasting, where recent fluctuations dominate. Guo et al. (2024) extend this idea with EEMD-CNN-GRU, decomposing the input series into trend, low-frequency, and high-frequency components, each modeled separately and then aggregated, highlighting the dominance of the trend component in driving predictive accuracy. In a similar vein, Song et al. (2023) apply VMD to separate financial signals into trend, periodic, and nonlinear disturbance components, where the linear components are assigned to ARIMA, and the nonlinear disturbances to LSTM. Similarly, Wen et al. (2024) adopt a multilevel wavelet decomposition (mWDN) approach to capture long-term and short-term cyclic patterns, addressing data leakage and boundary issues, which can skew forecasting performance. LSTM is then used for prediction.

Other approaches incorporate denoising techniques, such as DAE-LSTM, for detection of rising change points (Yoo et al., 2021) and wavelet denoising-enhanced hybrid LSTM and DB-ARIMA for smoothing data and revealing underlying trends (Ma et al., 2024).

Optimization-based architectures. Optimization-based architectures enhance DL models for stock prediction by improving convergence, feature selection, and parameter tuning. Studies such as Chen and Zhou (2020) use GA to rank and select the most influential financial and technical factors for LSTM input, while Kumar et al. (2022) introduce an Adaptive PSO-LSTM hybrid model that optimizes LSTM weights and biases, effectively addressing challenges such as overfitting and being stuck to local minima. Novel techniques leveraging PSO are introduced by Chauhan et al. (2023), who utilize PSO for hyperparameter tuning of LSTM and GRU models. The hybrid (PSO-LSTM) + (LSTM-GRU) ensemble outperforms in terms of accuracy, precision, recall, and F1-score. Similarly, Ji et al. (2021) propose the Improved PSO (IPSO)-LSTM model, which enhances optimization by incorporating an adaptive mutation factor, to avoid premature convergence to local optima, and a nonlinear approach to improve the PSO's inertia weight. Other works, such as Gülmez (2023) and Salahshour et al. (2024), leverage ARO's balance of exploration and exploitation to optimize LSTM performance.

Integrating technical and sentiment analysis. **Focus on Technical Analysis** Recent studies increasingly leverage technical indicators within hybrid DL frameworks to better capture stock market behavior. The CNN3D-DR+LSTM (Yang et al., 2020) converts technical indicators into deterministic trend signals and organize them in a 3D tensor using correlation-based ranking, enhancing CNN performance for better price prediction. The HW-LSTM-RL (Lee et al., 2021) use wavelet-denoised high-frequency components as input to an LSTM for reinforcement learning, focusing on optimizing trading strategies rather than direct prediction. Similarly, the Attention-enhanced Compound Neural Network (AComNN) (Yang et al., 2021) effectively models complex cross-regional (U.K., U.S., and China) financial dependencies using self-attention and a diverse set of technical and macroeconomic features, resulting in improved prediction accuracy and robust trading performance.

Hybrid architectures combining LSTM with ensemble methods, such as the LSTM-Forest model (Park et al., 2022), incorporates multi-task learning for stock return prediction and direction classification, with a multi-loss function combining MSE and cross-entropy. Cross-market models combining wavelet transforms with multi-input LSTM (Tang et al., 2024) utilize technical indicators and data from both Chinese and U.S. stock markets, leveraging inter-market correlations to improve forecasting accuracy. Additionally, the PCA-ICA-LSTM framework (Sarıkoç & Celik, 2024) employs PCA to reduce dimensionality and noise, followed by ICA for extracting independent features, thereby improving data quality and boosting return rates without sacrificing accuracy. Notably, Kalra et al. (2024) propose a novel hybrid bidirectional LSTM (H.BLSTM) model that combines incremental learning and DL techniques for real-time index price prediction. By leveraging univariate and multivariate time series data, including technical indicators, the model outperforms benchmark models across nine global stock indices with an average forecasting delay of just 2 s, making it ideal for real-time trading.

Focus on Sentiment Analysis

Li (2021) demonstrate that extracting deep emotional information via CNNs, rather than relying on basic emotional features, enhances predictive performance, particularly when integrating multiple information streams. Building on this, Liu et al. (2024b) propose SA-TrellisNet, a news-driven stock market index prediction model combining an LSTM-CNN-based sentiment analysis with a sentiment attention mechanism to dynamically weight stock price features based on their sentiment index, reflecting their significance. Meanwhile, Abdullah et al. (2024) construct sentiment indices (WHTCSI, HTCSI) using social media (Twitter) and Google Trends data (2006–2021) to forecast Halal tourism stock prices, showing strong predictive power via CNN over ARIMA, SVR, MLP, and LSTM models.

Focus on Technical - Sentiment Analysis

Integrating sentiment with technical indicators improves stock prediction, as shown by Prachyachuwong and Vateekul (2021), who combine LSTM and BERT on sector-segmented economic news headlines and price data. Similarly, Gangopadhyay and Majumder (2023) find that combining text and price data outperforms models based on price or text alone, with CNNs capturing price sequences well and Transformers, like BERT and FinBERT, excelling at text understanding.

Financial forecasting in crisis and pandemic contexts. Extreme events like the COVID-19 pandemic act as real-world stress tests, exposing the strengths and limitations of financial forecasting models under pressure. Exploring LSTM-based forecasting under COVID-19 conditions, Lachaab and Omri (2023) find KNN to outperform LSTM and ARIMA for CAC40 forecasts, highlighting the positive effect of containment and vaccination measures. In contrast, other works, such as Bathla et al. (2023) and Nazareth and Reddy (2024), confirm LSTM's strong performance on global and sector-specific indices, though challenges remain with extreme price fluctuations. Comparative studies like Ampountolas (2023) and Widiputra and Juwono (2024) highlight the advantages of hybrid and adaptive models in pandemic-era forecasting. Ampountolas (2023) show that the ETS-ANN model, blending linear (ETS) and nonlinear (ANN) components, consistently yields the lowest error rates across European indices and cryptocurrencies during pre-, mid-, and post-pandemic periods, despite some occasional drawbacks. Meanwhile, Widiputra and Juwono (2024) demonstrate that forecasting performance depends heavily on data characteristics: LSTM performs best with stable inputs, while CNN-LSTM better handles volatility. Vuković et al. (2024) examine market efficiency using LSTM, ARIMA, and LightGBM, revealing structural inefficiencies during crises, such as the 2007–2009 financial crisis and COVID-19 pandemic, challenging EMH assumptions and supporting the AMH framework. These findings underscore the importance of context-sensitive model selection and hybrid architectures that balance complexity with adaptability.

Advanced modeling techniques and approaches. Recent efforts focus on practical deployment and robustness. For example, Barra et al. (2020) employ CNN ensembles on Gramian Angular Fields images to extract spatial patterns for S&P500 trend prediction. Touzani and Douzi (2021) illustrate cross-market learning by training LSTM and GRU models on U.S. and French market data to predict (short- and medium-term) Moroccan stock prices, highlighting the potential of international data integration. Detecting and compensating for outliers, Naidoo and Du (2022) propose a two-stage DL framework, with an LSTM-based auto-encoder for anomaly detection, and an HONN to forecast stock prices. Wu et al. (2022) enhance standard DL models (MLP, RNN, LSTM, GRU) with transfer learning using industrial chain data, improving stock market index trend prediction accuracy, especially for MLP.

Both Wang, Lee (2023) and Aryal et al. (2020) underscore the increasing effectiveness of adopting multifractal analysis to better capture the complex and nonlinear dynamics inherent in financial markets. Wang, Lee (2023)'s neuro-inspired EINS model, combined with MultiFractal Asymmetric Detrended Cross-Correlation Analysis (MF-ADCCA), demonstrates notable improvements in volatility forecasting, suggesting that biologically motivated designs can enhance model adaptability to financial data intricacies. Complementing this, Aryal et al. (2020)'s comparative study highlights that while architectures like N-BEATS, ES-LSTM, and TCN excel in stock market forecasting, LSTM variants and Encoder-Decoder models are more suited to forex markets, reflecting the necessity of model selection tailored to asset class characteristics.

6.2.3. Summary

Recent advances in stock market index forecasting utilize DL models, with LSTM, CNN, and GRU being the most commonly used standalone architectures. Notably, hybrid models, particularly those combining CNN and LSTM, shows exemplary forecasting performance. However, due to the inherently noisy nature of financial data, hybrid approaches often incorporate signal decomposition techniques, tailored optimization, and integration of technical and sentiment features to further enhance prediction accuracy.

Different market components contribute differently to prediction accuracy, highlighting the importance of decomposition in forecasting. Techniques such as EMD and wavelet transforms isolate intrinsic signal components, enabling models to focus on more stable patterns and better handle market volatility and regime shifts. However, while denoising improves signal clarity, it risks smoothing out high-frequency patterns vital to short-term forecasting, requiring careful tuning. Additionally, given the often marginal gains over simpler models, the added complexity of decomposition methods should be carefully justified in practical applications.

Incorporating diverse data sources, such as technical indicators and sentiment analysis from social media and news, has demonstrated clear benefits. This multidimensional input enriches the feature space, offering a broader context for price movements and enhancing forecast accuracy. Nevertheless, challenges including overfitting, and real-time computational demands remain critical concerns.

Innovative approaches highlight the growing sophistication of DL architectures and the relevance of hybrid model design in modeling market anomalies and structural breaks. These studies underscore the practical adaptability of DL models and the value of domain-specific augmentation (e.g., outlier handling, transfer learning) for real-world forecasting. Integration of multifractal dynamics, neurodynamics, and sentiment analytics represents a promising direction for modeling complex, behavior-driven financial systems.

Cross-study observations. Across the reviewed studies ($n = 83$), standard LSTM and LSTM-based hybrid models are most frequently employed ($\approx 69\%$ combined), followed by integrations of technical and sentiment analysis ($\approx 14\%$, 12 studies), as well as optimization techniques, signal decomposition-denoising methods, and crisis or pandemic-focused approaches, each representing $\approx 7\%$ (6 studies).

6.3. Forex price forecasting

Foreign exchange (forex) markets are the largest in terms of trading volume, operating 24/7. With an average daily trading volume exceeding 7 trillion USD (BestBrokers, 2025), forex is the most liquid and heavily traded financial market worldwide. The rise of online trading platforms providing real-time market access to diverse options has driven the demand for profitable trading strategies. As a result, a significant body of research on forex forecasting using DL models has emerged. Given the dominance of the US Dollar in global transactions, particular emphasis is placed on the EUR/USD currency pair, which accounts for over 80% of the total forex trading volume (Yıldırım et al., 2021).

CNN-based architectures. Recent studies on forex price forecasting demonstrate the growing sophistication of hybrid DL architectures.

Hybrid approaches combining CNN-based candlestick image processing with CNN-LSTM time series analysis improve profitability by classifying moving average crossover signals into profitable long, short, or non-profitable trades, outperforming traditional benchmarks (Moghaddam & Momtazi, 2021). However, CNN-based image transformation methods are limited by their dependence on input quality and data volume, reducing robustness in market trend classification for trading (Tsai et al., 2019). More broadly, evidence from stock and forex markets suggests that CNNs generally underperform due to their limited capacity to effectively incorporate past data (Jing & Pathmanathan, 2023).

Fusions of LSTM and beyond. Alternatively, LSTM and transformer-based methods show strong forecasting capabilities. Integrating XGBoost for feature selection with LSTM temporal modeling enhances predictive accuracy over classical baselines such as ARIMA (Vuong et al., 2022). Novel dual-LSTM architectures that separately model macroeconomic and technical indicators improve directional prediction, though at the cost of increased complexity (Yıldırım et al., 2021). Evaluations across different asset classes further show the strong performance of LSTM variants and encoder–decoder architectures in forex forecasting (Aryal et al., 2020), while hybrid CNN-LSTM models have demonstrated effectiveness across stocks, forex, and cryptocurrencies (Abdullah & Salah, 2023). Transformer architectures with time embeddings demonstrate promising results in short-term intraday forecasting, outperforming LSTM and ARIMA models, especially when incorporating multivariate inputs (Fischer et al., 2024).

Table 4
Cross-study overview of index forecasting: Models, assets, features, and timeframes.

Category	Top Entities/Statistics
Most Common Models	Standalone: LSTM (25), CNN (9), GRU (9); Hybrids: LSTM-based (32), CNN-LSTM-based (8). Total applications: 131[a b].
Standalone vs. Hybrid Use	Standalone: ~78%; Hybrid/ensemble models: ~77%; Novel architectures: ~31%.[c]
Frequent Feature Types	Price-based features (e.g., OHLCV, Close, Adjusted Close) appear in most studies (≈ 92%); technical indicators appear in ≈ 22% of cases; sentiment features (e.g., news, tweets) in only ≈ 6%. Notably, ≈ 28% (23 studies) rely solely on Close price data. Rarely used features: chaotic-modeled time series, dividend yields, GAF images.
Most Forecasted Stock Indices	U.S. indices dominate, led by S&P500 (~37%), DJI (~18%), and NASDAQ (~10%). Chinese indices follow with SSE (~19%) and CSI300 (~11%), along with HSI from Hong Kong (~8%) and Nifty50 from India (~7%).
Geographic Focus	Global or mixed-region studies dominate (~40%, 33 studies), followed by the single-study cases U.S. (~22%, 18 studies) and China (~18%, 15 studies). Other regional focuses include India (~6%, 5 studies), Europe (~4%, 3 studies), and Pakistan, Saudi Arabia, South Korea, and Thailand (each ~1%).
Time Periods Covered	Most studies use long-term datasets over 10 years (~70%, 58 studies), while fewer focus on periods <5 years (~28%, 23 studies). The earliest dataset begins in 1983, with many extending into the early 2020s. The 2010–2021 decade is the most studied (~20%, 17 studies). Some analyze multiple periods, but overall, longer historical datasets are preferred.
Performance Metrics Used	Most commonly reported evaluation metrics: RMSE (~58%, 48 studies), MAE (~54%, 45), MAPE (~49%, 41), and MSE (~37%, 31). R^2 appears in ~23% (19 studies). Classification-oriented metrics such as Accuracy (~19%, 16), F1-Score (~10%, 8), Precision (~10%, 8), and Recall (~8%, 7) are less frequent. A few studies also report Theil's U (5), and SMAPE (4).

[a] Some studies use multiple models (e.g., baselines, ensembles), contributing to total model count > number of papers (83). This reflects broader comparative experimentation.

[b] Model groups are defined as follows: LSTM-based (e.g., LSTM-ARIMA, PSO-LSTM, VAR-LSTM); CNN-LSTM-based (e.g., CNN-LSTM, CNN–LSTM–Transformer, CNN-BiLSTM).

[c] Percentages represent the ratio of the total number of model appearances to the 83 studies reviewed. Note that individual studies may include multiple models, including both standalone and hybrid types.

Chaos-based advanced modeling techniques. It is noteworthy that novel chaos-based hybrid models, extensively discussed in index forecasting literature, address nonlinear, noisy, and chaotic market behaviors by integrating chaos theory with DL models and post-processing refinements such as polynomial regression (Durairaj & Mohan, 2021, 2022; Durairaj et al., 2023). These approaches consistently outperform traditional statistical and DL models, emphasizing the value of domain-specific augmentations to better capture financial time series complexities. To avoid redundancy, detailed discussion of these chaos-based models is presented in the index forecasting section, where their methodological innovations and empirical superiority are thoroughly described.

6.3.1. Summary

The literature on DL-based forex forecasting reveals a strong trend toward hybrid and multivariate modeling frameworks. CNN-based models, particularly those leveraging candlestick chart transformations, offer a promising approach to signal classification. However, their performance is often hindered by sensitivity to input quality and an inherent limitation in capturing temporal dependencies (Jing & Pathmanathan, 2023; Moghaddam & Momtazi, 2021; Tsai et al., 2019). These findings align with a broader skepticism regarding CNNs for financial time series prediction, especially when past temporal context is crucial.

In contrast, LSTM architectures, especially those enhanced through feature selection (e.g., XGBoost) or hybridized with macroeconomic and technical indicators, consistently demonstrate strong predictive performance (Vuong et al., 2022; Yıldırım et al., 2021). Transformer-based models with time embeddings have emerged as a compelling alternative, particularly for high-frequency, short-horizon forecasting tasks (Fischer et al., 2024). My impression is that transformers are underutilized in forex relative to their potential, especially in scenarios requiring attention to multiple time scales and cross-asset dependencies.

The introduction of chaos-theory-enhanced models presents an intriguing deviation from conventional DL approaches. Evaluated under index forecasting contexts, their success in capturing market nonlinearity and noise suggests significant untapped potential in forex prediction as well (Durairaj & Mohan, 2021, 2022; Durairaj et al., 2023).

Cross-study observations. Across the 11 reviewed studies, LSTM models, both standalone and hybrid, are predominant, appearing on average 1.18 times per study, indicating that each study uses at least one LSTM variant. Technical analysis features (≈ 36%, 4 studies) and chaos-based models (3) are also commonly explored in forex forecasting. Notably, the US dollar consistently serves as the primary base currency, highlighting its dominant role in this research area.

Table 5
Cross-study overview of forex forecasting: Models, assets, features, and timeframes.

Category	Top Entities/Statistics
Most Common Models	Standalone: LSTM (6), CNN (4); Hybrids: LSTM-based (7). Total applications: 27[a b].
Standalone vs. Hybrid Use	Standalone: ∼154%; Hybrid/ensemble models: ∼90%; Novel architectures: ∼36%.[c]
Frequent Feature Types	Price-based features (e.g., OHLC, Open, Close) appear in most studies (≈ 90%, 10 studies); close price data and technical indicators each appear in ≈ 36% (4 studies). Rarely used features include chaotic-modeled time series and macroeconomic indicators.
Most Forecasted Forex Currency Pairs	EUR/USD leads (≈ 55%, 6 studies); INR/USD, JPY/USD, and SGD/USD, follow, each appearing ≈ 27% (3 studies). Notably, the US dollar serves as the primary base currency in forex forecasting studies, reflecting its global market dominance.
Time Periods Covered	Several studies use long-term datasets over 10 years (∼45%, 5 studies), while fewer cover periods under 5 years (∼27%, 3 studies). The longest dataset spans 32 years (1990–2021) and appears in two studies.
Performance Metrics Used	The most frequently reported evaluation metrics are MSE (∼45%, 5 studies), RMSE and MAE (each ∼36%, 4), followed by MAPE, D-statistic, and Theil's U (each ∼27%, 3). Less commonly employed metrics include classification-based measures as well as trading-specific indicators such as the number of transactions and profit accuracy.

[a] Some studies use multiple models (e.g., baselines, ensembles), contributing to total model count > number of papers (11). This reflects broader comparative experimentation.

[b] LSTM-based models include Chaos-LSTM-PR, ES-LSTM, Stacked-LSTM, and XGBoost-LSTM.

[c] Percentages represent the ratio of total model appearances (standalone, hybrid, novel) to the number of studies (11), reflecting how frequently models appear per study on average. For example, 154% means standalone models appeared on average 1.54 times per study, implying each study uses at least one standalone model. Note that individual studies may include multiple standalone and/or hybrid models.

6.4. Commodity price forecasting

Commodity forecasting predicts the future prices of physical goods like oil, gold, agricultural products, and metals. Given their key role in global trade and sensitivity to factors such as supply and demand, geopolitical events, and economic conditions, accurate forecasting is vital for investors, businesses, and policymakers.

Leveraging recurrent architectures. Recent studies leverage recurrent neural networks to enhance predictive performance. Halim et al. (2022) demonstrate that an attention-enhanced multivariate LSTM outperforms baseline models (RNN, CNN, Transformer) in forecasting prices of 11 key Indonesian food commodities. In a related effort, Huang et al. (2023) propose a CNN-GRU hybrid augmented with PCA for dimensionality reduction. Their findings on copper and soybean meal futures reveal that hybridization not only enhances prediction accuracy but also reduces computational cost, compared to individual CNN or GRU models, a valuable trait for real-time forecasting in commodity markets.

Integrating reinforcement learning and interpretability. Incorporating learning agents, Tong and Yin (2022) and Ansari et al. (2022) explore reinforcement learning-based systems for commodity trading. The former employs LSTM in a high-frequency trading setup for corn futures, outperforming a buy-and-hold strategy in risk-adjusted returns. The latter proposes a novel DRL-based decision support system integrating GRU forecasting with reinforcement learning. By enhancing the agent's observation state with both historical and predicted trends, this system enables fast training, with profitable trading decisions across multiple markets. These approaches underscore the potential of reinforcement learning to move beyond passive forecasting toward active decision-making, a shift that aligns well with the growing emphasis on autonomous financial systems.

More recently, Wu et al. (2024) contribute a novel interpretable forecasting framework for corn futures price prediction, that marries signal decomposition (CEEMDAN), CNN-based text mining, and an optimized Temporal Fusion Transformer (TFT) via adaptive differential evolution with optional external archiving (JADE). The model's use of multisource heterogeneous inputs, including supply–demand dynamics, policy changes, geopolitical risks, and market sentiment, resulting in highly accurate and interpretable weekly price forecasts, marks a step forward in modeling the complex drivers of commodity prices. Importantly, their emphasis on interpretability adds transparency to the modeling process, an often-overlooked but increasingly vital requirement in financial applications.

Chaos-based advanced modeling techniques. Finally, chaos-based methods, extensively discussed in the index forecasting section, also show promise in commodity markets, in addition to forex. The Chaos-LSTM-PR model by Durairaj and Mohan (2021) and its subsequent extensions (Durairaj & Mohan, 2022; Durairaj et al., 2023) consistently outperform traditional approaches and chaos-augmented models across crude oil, gold, and soybeans. These results suggest that capturing chaotic structures may be particularly useful in markets characterized by nonlinearity and structural breaks, common features in commodity time series, and other asset classes.

In our view, while LSTM and hybrid DL models remain dominant, emerging approaches that emphasize decision support, interpretability, and heterogeneous data integration represent the most promising directions. The integration of reinforcement learning and decomposition techniques, in particular, signals a move toward more robust, adaptive, and explainable forecasting frameworks that better reflect the real-world complexity of commodity markets.

6.5. Bond price forecasting

Bond forecasting involves predicting the future prices or yields of debt securities issued by governments or corporations, with some experts arguing that bond prices better reflect economic health than the stock market. Only two studies were identified in our review that focus explicitly in this field. Both Xu, Su, et al. (2024) and Ping et al. (2024) emphasize the critical role of feature dimensionality reduction techniques, such as LASSO and PCA, in enhancing predictive performance when combined with LSTM architecture. Xu, Su, et al. (2024) propose a hybrid LASSO-PCA-LSTM model to predict the monthly return rate of the Securities Bank of China, demonstrating that this combination yields superior accuracy than similar models that use GRU or RNN in place of LSTM. Similarly, Ping et al. (2024) present a more elaborate framework integrating LASSO, SMLR, PCA, and Bayesian optimization with LSTM to forecast treasury bond yields. Their systematic evaluation shows that stepwise integration of these techniques incrementally improves model performance, with the LASSO-SMLR-PCA-LSTM configuration outperforming a wide range of baseline models, including BPNN, CNN, RNN, Transformer, and several other LSTM-based configurations such as LASSO-LSTM, PCA-LSTM, LASSO-SMLR-LSTM, and LASSO-PCA-LSTM.

These findings highlight the utility of integrating feature selection, dimensionality reduction, and recurrent networks, especially LSTM, in bond price prediction, though the limited research to date limits broad conclusions. In our view, these studies primarily reinforce LSTM's strong predictive capabilities, even in more stable asset classes like bonds, with specific enhancements further strengthening performance. Looking ahead, future research should aim to explore emerging architectures like ESNs, and enrich input data with alternative sources such as inflation expectations, employment statistics, and ESG-related indicators, to better capture bonds' sensitivity to macroeconomic and policy shifts, especially in the context of green finance.

6.6. Cryptocurrency price forecasting

Cryptocurrencies, as peer-to-peer digital currencies, rely on blockchain technology to securely record transactions within individual blocks (Kang et al., 2022). The decentralized nature of blockchain provides advantages such as low transaction costs, seamless circulation, and enhanced security (Li et al., 2022). Bitcoin, introduced by Nakamoto (2008) as the first decentralized cryptocurrency, has since evolved into the most valuable digital currency worldwide.

GRU-based forecasting. Several studies underscore the effectiveness of GRU models in cryptocurrency price prediction. GRU has been shown to outperform both LSTM and BiLSTM in forecasting the prices of cryptocurrencies such as Bitcoin, Ethereum, and Cardano (Aljadani, 2022; Dutta et al., 2020), and it also demonstrates superior accuracy compared to LSTM, CNN, and CNN-LSTM models in predicting price fluctuations of the NEAR cryptocurrency (Gollapalli et al., 2023). This suggests GRU's robustness across diverse crypto assets. However, Al-Nefaie and Aldhyani (2022a) challenge the dominance of GRU by showing that an MLP yields better predictive accuracy for Bitcoin, indicating that non-recurrent models may be competitive when features are carefully engineered or when temporal dependencies are minimal.

LSTM and temporal modeling. Despite GRU's strong performance, LSTM networks remain prominent in forecasting cryptocurrencies such as Bitcoin, Ethereum, and Dogecoin, consistently outperforming other RNN variants, CNN models (Nair et al., 2023), and traditional methods like ARIMA (Fleischer et al., 2022), largely due to their superior ability to capture long-range temporal dependencies. High-frequency data further validate LSTM's strengths. Bouslimi et al. (2024) analyze minute-level Bitcoin and Ethereum prices during COVID-19, revealing that LSTM, particularly for Ethereum, maintains high accuracy in volatile environments.

Alternative models: FB-prophet and ARIMA. In contrast to DL models, Tripathy et al. (2023) evaluate FB-Prophet, ARIMA, and LSTM, reporting that FB-Prophet achieves the highest predictive accuracy for Bitcoin. This result is noteworthy given FB-Prophet's simplicity and interpretability, highlighting its utility in scenarios with strong seasonality or trend components. Mahjouby et al. (2024) conduct a broader comparison involving linear regression, SGDRegressor, random forest, AdaBoost, LSTM, and ARIMA on CoinMarketCap and CryptoCurrency datasets. The study finds that ensemble and linear models (e.g., random forest regressor and linear regression) often outperform DL models. Although ARIMA performs competitively, the study ultimately advocates for a hybrid approach that combines multiple models to leverage their complementary strengths.

Hybrid architectures and advanced modeling techniques. Hybrid DL models have shown notable improvements in cryptocurrency forecasting by combining the strengths of different architectures. For instance, the 1dCNN-GRU model (Kang et al., 2022) effectively integrates CNN's pattern recognition with GRU's temporal modeling, outperforming various benchmarks across Bitcoin, Ethereum, and Ripple cryptocurrencies. Similarly, the LSTM-GRU hybrid (Patel et al., 2020) improves short-term prediction accuracy over LSTM-based models for forecasting Litecoin and Monero prices. Other hybrids like GRU-BiLSTM (Ferdiansyah et al., 2023) and CNN-LSTM (Abdullah & Salah, 2023) extend these gains across diverse assets and markets. Additionally, the PSO-optimized Improved Deep Belief Network (IDBN) (Li et al., 2022) exemplifies how combining neural architectures with optimization techniques can further boost accuracy.

In the context of the natural stress-testing scenario brought about by the COVID-19 pandemic, a comparative analysis by Ampountolas (2023) reveals that hybrid models like ETS-ANN outperform traditional parametric (ARIMA) and non-parametric (kNN) approaches in forecasting both stock market indices and cryptocurrency (BTC/USD) prices. Despite this improvement, the study highlights the persistent challenge of achieving high predictive accuracy in volatile financial markets, where forecasting performance often remains limited. To address these limitations, ensemble learning techniques have shown strong potential in enhancing predictive performance. Livieris et al. (2020) demonstrate that combining DL architectures, such as LSTM, BiLSTM, and convolutional layers, within ensemble frameworks (e.g., averaging, bagging, stacking) consistently outperforms individual models for cryptocurrency prices and directional movement prediction. Notably, stacking with a kNN meta-learner yields superior results, suggesting that combining diverse model perspectives can better capture market complexity. Further extending the frontier, generative models like TimeGAN (Santoro & Grilli, 2022) provide valuable insights into how market temporal characteristics affect predictive performance. Their analysis suggests that cryptocurrency markets, due to their continuous 24/7 nature, possess stronger discriminative and predictive power compared to traditional time-constrained stock markets. Incorporating additional features such as yield can further enhance the predictive power of stocks, indicating the potential for hybridizing generative and other DL approaches with feature engineering.

Novel hybrid approaches. Several recent studies have introduced innovative approaches to cryptocurrency forecasting. For example, Ahmad et al. (2024) propose a CNN-LSTM-Transformer model that leverages self-attention alongside convolutional and recurrent layers, achieving superior performance on extensive minute-level historical data of various cryptoassets dating back to the 1970s, as well as on international stock indices, albeit with increased computational demands. Similarly, Jin and Li (2023) present the VMD-AGRU-RESVMD-LSTM hybrid model, which uses VMD and EMD for multi-stage decomposition, attention-enhanced GRUs for subsequences prediction, and LSTM for integration, outperforming both standalone and other hybrid models in Bitcoin and Ethereum forecasting.

Efforts to model more specialized phenomena include Kallurkar and Chandavarkar (2024), who focus on Ethereum transaction fee prediction post-EIP-1559,[1] using a CNN-LSTM model to capture local and temporal features, thereby addressing a notable gap in cryptocurrency transaction forecasting. The Multiple-Input Convolution DL (MICDL) model by Livieris et al. (2021) further extends hybridization by processing multiple cryptocurrencies independently through convolutional and recurrent layers, improving both regression and classification tasks.

Other studies focus on the integration of alternative data sources to enrich forecasting models. In particular, the use of sentiment data in cryptocurrency prediction, as shown by Patel et al. (2024) and Jung et al. (2024), highlights the increasing importance of information beyond traditional price series. Both studies leverage BiLSTM-based architectures to effectively integrate heterogeneous information, such as social media sentiment from platforms like Twitter, technical indicators, on-chain transaction metrics, and broader macroeconomic variables. Patel et al. (2024) focus on combining hourly Bitcoin price data with sentiment extracted from social media, showing that models enriched with sentiment signals significantly outperform various LSTM variants and price-only models. Similarly, Jung et al. (2024) extend this approach by incorporating a wider array of data categories, ranging from news and social media sentiment scores generated by VADER and FinBERT, Google Trends insights, to on-chain and financial asset data, evaluating multiple algorithms and demonstrating that BiLSTM models particularly excel in capturing short-term price movements with a three-day window. Their findings demonstrate that the fusion of sentiment, technical, and on-chain data outperforms price-only models. A key strength of their study is the extensive dataset, which spans over 2000 days and encompasses more than five distinct categories of information, offering a wider and more detailed forecasting framework. These advances reflect a broader trend in the field toward hybrid modeling strategies that incorporate diverse and complementary data sources.

6.6.1. Summary

Overall, the literature reveals no universal best model for cryptocurrency forecasting. Model performance varies significantly based on the target cryptocurrency, data granularity, and feature selection. GRU and LSTM are consistently strong contenders. However, simpler models like MLP and FB-Prophet can outperform deep architectures in certain contexts, suggesting that model selection should be guided by empirical validation rather than architectural complexity. Furthermore, hybrid and ensemble methods represent a promising direction, with novel studies incorporating attention mechanisms, time series decomposition techniques, and sentiment analysis to enhance predictive accuracy. Notably, the majority of forecasting analyses concentrate on the post-2015 period, reflecting the market's transition into a more mature and data-rich phase, and Bitcoin is the most extensively studied asset.

Cross-study observations. Across the reviewed studies (n = 23), standard LSTM and ML models are most frequently employed ($\approx 96\%$ combined) in terms of model appearances per study; technical and sentiment analysis are incorporated in 3 studies ($\approx 13\%$). Price-based features like OHLCV are universally utilized, though more specialized inputs such as on-chain data and hash rate are less common.

6.7. Volatility forecasting

Volatility forecasting plays a central role in risk management and asset pricing. It enables investors to better anticipate market fluctuations, optimize portfolios, and make informed decisions under uncertainty.

[1] Ethereum Improvement Proposal 1559 (EIP-1559), part of the Ethereum London upgrade, restructured the fee mechanism by introducing a base fee and an optional tip, aiming to reduce congestion and improve transaction confirmation times.

Table 6
Cross-study overview of cryptocurrency forecasting: Models, assets, features, and timeframes.

Category	Top Entities/Statistics
Most Common Models	Standalone: LSTM (12), ML (10), GRU (6); Hybrids: CNN-LSTM-based (4). Total applications: 53[a b].
Standalone vs. Hybrid Use	Standalone: ∼178%; Hybrid/ensemble models: ∼48%; Novel architectures: ∼26%.[c]
Frequent Feature Types	Price-based features (e.g., OHLCV, Close, Adjusted Close) are used in all studies ($\approx 113\%$), with some relying exclusively on OHLCV and others combining it with Adjusted Close. Less common features include on-chain data metrics, hash rate[d], and volatility.
Most Forecasted Crypto Assets	Bitcoin dominates (∼78%, 18 studies), followed by Ethereum (∼39%, 9) and Ripple (∼17%, 4).
Time Periods Covered	Nearly half of the studies (∼48%, 11 studies) use short-term datasets under five years, while only a few (∼13%, 3) span ten years or more, with the longest covering 11 years. Most focus on the post-2015 period, reflecting the maturing phase of cryptocurrency markets.
Performance Metrics Used	The most commonly reported evaluation metrics include RMSE (∼83%, 19 studies), MAE (∼57%, 13), MSE (∼39%, 9), and MAPE (∼30%, 7). R^2 appears in ∼35% (8 studies). Less common measures include classification-oriented metrics, and statistical or information-theoretic metrics such as Kullback–Leibler divergence and t-SNE.

[a] Some studies use multiple models (e.g., baselines, ensembles), contributing to total model count > number of papers (23). This reflects broader comparative experimentation.

[b] CNN-LSTM-based models include CNN-LSTM, and CNN–LSTM–Transformer.

[c] Percentages represent the ratio of total model appearances (standalone, hybrid, novel) to the number of studies (23), reflecting how frequently models appear per study on average. For example, 178% means standalone models appeared on average 1.78 times per study, implying that most studies employ nearly two standalone models rather than just one. Note that individual studies may include multiple standalone and/or hybrid models.

[d] Hash rate refers to the computational power used per second in cryptocurrency mining.

Standalone LSTM for market volatility. Recent studies in volatility forecasting show the advantages of DL models, especially LSTM, over traditional methods like GARCH. Wang, Andreeva, and Martin-Barragan (2023) examine the predictive power of internal factors (e.g., lagged volatility, moving averages) and external factors (e.g., technology, financial, and policy uncertainty data) using LSTM and RF models, finding them superior to traditional volatility models like GARCH. The study highlights that GA- and ABC-optimized LSTMs, particularly ABC-LSTM, offer the highest accuracy for weekly volatility predictions. Additionally, a universal model trained on multiple cryptocurrencies outperforms cryptocurrency-specific models, reflecting the interdependent nature of the market. Similarly, Al-Khasawneh et al. (2024) find LSTM models to be more effective than traditional approaches such as GARCH and TARCH in capturing the complex, nonlinear patterns characteristic of stock index volatility on the Pakistan Stock Exchange (2009–2021). Their use of multivariate BiLSTM architectures yields the highest forecasting accuracy, underlining the benefit of exploiting temporal and cross-variable dependencies. Extending the application to broader equity markets, Petrozziello et al. (2022) explore both univariate and multivariate LSTM models for volatility forecasting using datasets from the DJI (2002–2008) and NASDAQ 100 (2012–2017). Their contribution lies in constructing a nonparametric, market-wide volatility model that leverages LSTM's strength in capturing cross-asset temporal dependencies. Their results confirm LSTM's superiority over traditional models such as R-GARCH and GJR-MEM, particularly under high-volatility regimes, as well as over RNN counterparts.

Hybrid architectures for stock volatility. Hybrid approaches to forecasting stock volatility show promising advancements by effectively combining the strengths of diverse models. For instance, Liu et al. (2023) introduce the Ordinal Regression GAN for Financial Volatility Trends (ORGAN-FVT), which uses ConvLSTM as a generator to learn the data distribution and an MLP as a discriminator to classify volatility trends in stock data. Enhanced with ordinal regression, ORGAN-FVT outperforms LSTM, GRU, CNN, and ConvLSTM across multiple stock datasets using both stock prices and technical indicators. This highlights the added value of combining generative modeling with trend-aware classification for volatility forecasting.

Several studies propose novel approaches. For example, Akter et al. (2022) focus on frontier markets, which often lack clear volatility parameters for risk assessment. They develop a stacking ensemble that combines CNN, LSTM, and BiLSTM base learners with a linear regression meta-learner. This model outperforms traditional methods like Random Forest, AdaBoost, and Gradient Boosting in forecasting market risk in Bangladesh. Their results suggest that ensemble hybridization is an effective strategy to overcome data limitations and handle the complexities of volatility in emerging economies. Expanding on these efforts, Araya et al. (2024) propose an ambitious multi-component hybrid model (SARIMA-GARCH-CNN-BiLSTM) that elegantly unites classical time series decomposition, volatility modeling, and deep feature extraction. While conceptually compelling, the resulting complexity poses valid concerns regarding scalability and applicability in real-time trading environments, where computational efficiency is critical. Similarly, Jin and Zhang (2024)'s integration of Wavelet Transform, CNN, and Transformer achieves impressive results in financial risk prediction, particularly benefiting from multi-scale feature extraction and attention mechanisms. However, the sensitivity to extreme outliers and increased computational complexity underscore a key challenge: balancing predictive power with robustness and operational feasibility.

Hybrid architectures for index volatility. In the domain of index volatility, Di-Giorgi et al. (2023) integrate GARCH with RNN variants (LSTM, BiLSTM, GRU) to capture heteroscedasticity and temporal dependencies. Empirical evaluations across global indices (IPSA, S&P 500, ASX200) confirm that BiLSTM-enhanced hybrids yield the most accurate short-term returns volatility forecasts. These results support the view that GARCH-DL integration provides a flexible and effective approach to modeling volatility structures. Complementing this, Jung and Choi (2021) present a novel autoencoder-LSTM hybrid for forecasting FXVIX indices, specifically EUVIX, BPVIX, and JYVIX (2010–2019). Subperiod analysis around global economic events such as the subprime mortgage crisis and Brexit, shows that incorporating wider data distributions and outliers improves forecasting accuracy. The model's ability to capture data variability and singularities makes it particularly well-suited for the highly reactive nature of currency volatility.

Hybrid architectures to commodity volatility. Lastly, Vidal and Kristjanpoller (2020) explore volatility forecasting in commodity markets, proposing a CNN-LSTM model that transforms price time series into image-like representations for deep feature extraction. Applied to five decades of gold price data, the model outperforms traditional and DL baselines (GARCH, SVR, LSTM, CNN), with Model Confidence Set (MCS) analysis confirming its superior predictive performance. This highlights the potential of hybrid architectures in capturing the multifaceted nature of commodity volatility directly shaped by economic, geopolitical, and market-driven factors.

Recent advances in volatility forecasting highlight the effectiveness of LSTM both as a standalone model and within hybrid frameworks. Approaches that integrate GARCH with RNNs or combine wavelet decomposition with Transformers push the boundaries by leveraging complementary model strengths. From generative models like ORGAN-FVT to modular ensembles for frontier markets, the evolution reflects not just a race for accuracy, but a deeper attempt to model financial complexity in a more interpretable and robust way. That said, a recurring challenge emerges: the trade-off between model sophistication and practical deployability.

6.8. Other works

Several recent studies extend the application of DL techniques in financial forecasting from different perspectives.

Addressing key limitations of DL models like their black-box nature, tendency to overfit, sensitivity to parameter choices and noise, slow convergence, and high computational demands, Zhang et al. (2021) propose obtaining prior knowledge of the correlation between historical and future stock prices. They introduce two feature extraction modes: the curve-shape-feature (CSF), in which future prices can be inferred from the curve shapes of historical price patterns, and the non-curve-shape-feature (NCSF), where such information is not encoded in price curve patterns. Their findings reveal that while a wide range of models, including SVM, RF, MLP, Bayesian classifiers, decision trees, CNN, embedded CNN, BiLSTM, and embedded BiLSTM, perform well under the CSF mode, they struggle under the NCSF mode. Importantly, real-world stock data suggests that CSF-mode contains limited predictive information, whereas NCSF-mode may offer stronger signals. This distinction suggests that much of the apparent success of existing models may be tied to their performance on datasets that do not reflect real market complexity, and highlights the need for novel modeling approaches capable of effectively capturing NCSF-mode for improved financial forecasting.

In the context of portfolio management, Ma et al. (2021) integrate a variety of ML and DL forecasting models, such as RF, SVR, LSTM, and CNN, and the statistical ARIMA with two portfolio optimization techniques: Mean–Variance Framework (MVF) and Omega Function (OF). Their results, based on CSI 100 data from 2007–2015, indicate that RF delivers the best predictive performance among all methods. When integrated with MVF, RF yields the most profitable portfolios even after accounting for transaction costs, although high turnover reduces profitability significantly. Interestingly, SVR+OF surpasses RF+OF when costs are included. Notably, this is the first study to combine ML and DL methods with MV and Omega portfolio optimization. The study highlights the critical importance of incorporating practical constraints, such as turnover and costs, which are often overlooked in purely predictive exercises. It also reinforces the robustness of RF in financial applications.

Addressing portfolio optimization complexities and return forecasting challenges, Wang, Wu, and Wu (2024) propose a hybrid model that combines LSTM networks for stock returns' temporal feature extraction, with a stochastic dominance filter for input dimensionality reduction. The model is designed to construct an index-tracking portfolio that replicates the performance of a specific market index by mirroring its composition and returns, instead of actively selecting individual stocks. By extracting dominant returns from 100 stocks to track the FTSE 100 index return, experiments demonstrate that the model improves prediction accuracy and reduces estimation errors. The integrative use of stochastic dominance for feature selection is particularly noteworthy, as it aligns economic theory with ML practice. While the results are promising, key concerns remain regarding the model's scalability, robustness across different indices, and practical implementation, particularly in terms of transaction costs. A comparison with conventional index-tracking methods would also help contextualize its relative performance and interpretability.

Due to its use of specialized, non-standard datasets, the final study included is the one by Zhao et al. (2024), who propose an advanced hybrid DL model for stock market prediction and investment decision-making, integrating CNN, BiLSTM, and an attention mechanism. The model utilizes CNN for feature extraction from historical stock prices and trading volumes, BiLSTM to capture temporal dependencies, and the attention layer to prioritize relevant features. Evaluation on datasets enriched with sentiment and market indicators (e.g., CAMBRIA, KRIRAN, SHARMA, and JAMES) shows that the CNN-BiLSTM-attention model outperforms state-of-the-art methods across multiple metrics. This work highlights the value of hybrid and attention-based architectures in complex, noisy environments like financial markets.

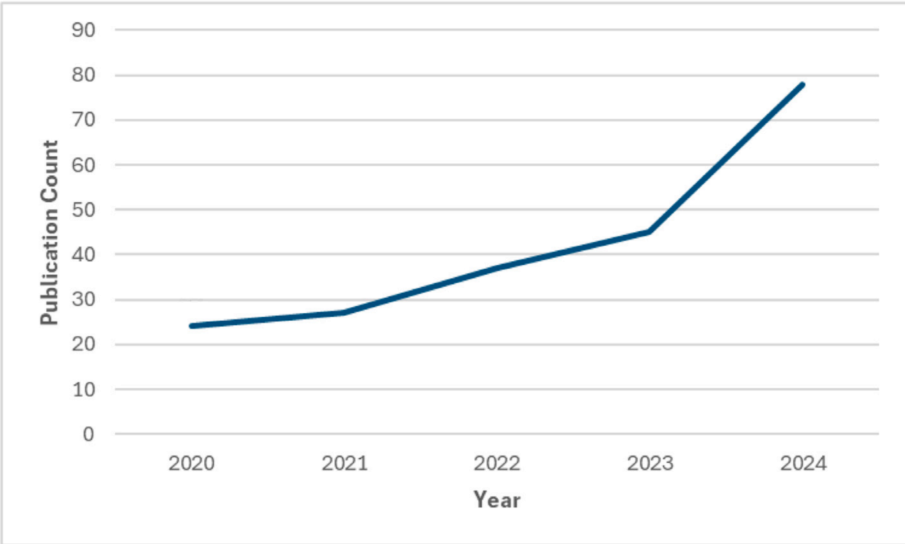

Fig. 11. Annual distribution of publication count (2020–2024).

Table 7
Summary of deep learning model performance across financial forecasting domains.

Domain	# Studies	Common hybrids and notes
Stock	75	Standalone LSTM models dominate. Price-based features are most common; sentiment and technical indicators moderately used; datasets mostly long-term.
Index	83	LSTM-based hybrids dominate. Mostly price features; cross-market studies focus on US and Chinese indices; mixed standalone and hybrid use; includes chaos-based modeling; mostly long-term data.
Forex	11	Standalone LSTM or LSTM-based hybrids. Major currency pairs centered on USD; mix of short- and long-term periods; includes chaos-based modeling.
Commodities	8	LSTM hybrids. Underexplored domain; hybrids improve accuracy and efficiency; emerging reinforcement learning and interpretability methods; includes chaos-based modeling.
Bond	2	LASSO-PCA-LSTM hybrids. Very limited research; feature selection key; considered a stable asset class.
Cryptocurrency	23	LSTM models. Shorter timeframes common, especially post-2015; Bitcoin and Ethereum dominate; unique features like on-chain data.
Volatility	10	Standalone and hybrid LSTM models, outperforming traditional models (e.g., GARCH).

7. Current snapshot of DL research for financial applications

After reviewing research articles specifically focused on financial forecasting with DL models, we present an overview of the current state of the field. A total of 187 papers were analyzed, and categorized based on the forecasted asset type. Additionally, we analyzed these studies in terms of their chosen DL models, development environment, datasets, and other distinguishing factors such as feature sets and performance evaluation metrics. Given that some articles fall under multiple forecasted asset types and employ various model types, the findings provide a general overview of the prevailing trends rather than a one-to-one mapping. For instance, the work of Durairaj and Mohan (2021) appears in three categories: index, forex, and commodity forecasting. A detailed breakdown of the reviewed studies is presented in the Online Appendix (available at: Download here), where the literature is systematically organized by asset class, including stocks, indices, foreign exchange, commodities, bonds, cryptocurrencies, and volatility, highlighting key characteristics such as datasets, time periods, feature sets, model architectures, evaluation metrics, and implementation environments.

The literature suggests that no single model is universally best, with optimal performance depending on task specificity, asset volatility, data characteristics, and the incorporation of domain knowledge. The following table provides key insights into DL model performance across multiple financial domains, highlighting research volume, common model types, and key observations within each domain (see Table 7).

We now present a summary of our key findings, emphasizing important insights for researchers in the field. Fig. 11 shows the annual distribution of published articles from 2020 to 2024, while Fig. 12 illustrates the allocation of research across different financial forecasting domains. The analysis reveals that index forecasting is the most extensively studied area, followed closely by stock price forecasting, with 83 and 75 studies, respectively. Together, these two domains represent over 70% of all reviewed research, underscoring their prominence in the literature. Additionally, cryptocurrency price prediction attracts considerable

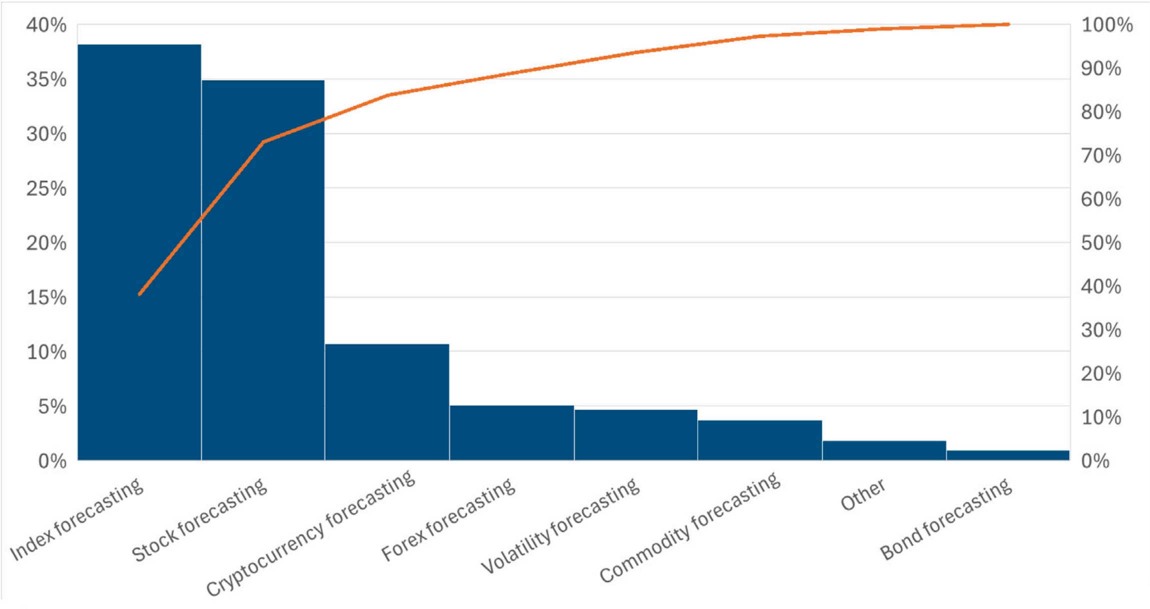

Fig. 12. Distribution of studies by financial forecasting task (2020–2024).

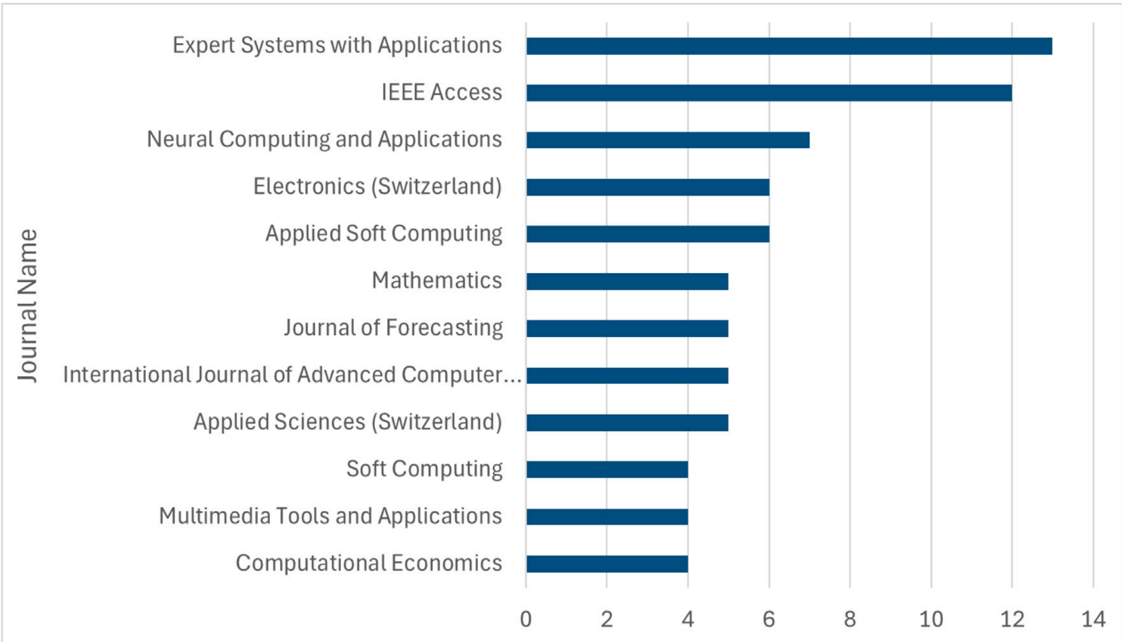

Fig. 13. Top journals by article publication count for the period 2020–2024.

attention with 23 studies, while forex forecasting accounts for 11 studies, and market volatility forecasting for 10. Commodity forecasting is less explored, with 8 studies, and bond market forecasting remains the least studied, with only two publications to date.

A key consideration for researchers is where they can publish their findings. As part of this review, we analyzed the distribution of publications across academic journals. The results, presented in Fig. 13, highlight the leading journals for financial time series forecasting. The journals with the most published articles include Expert Systems with Applications. IEEE Access, Neural Computing and Applications, Electronics (Switzerland), Applied Soft Computing, Mathematics, Journal of Forecasting, International Journal of

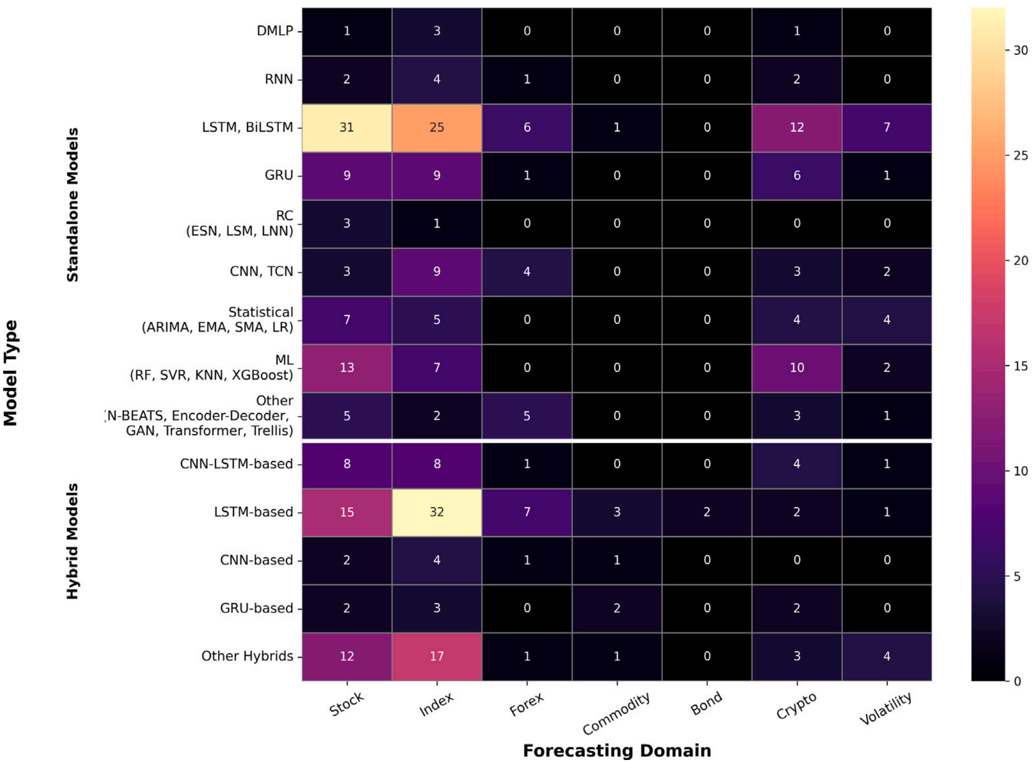

Fig. 14. Topic-Model heatmap.

Advanced Computer Science and Applications, Applied Sciences (Switzerland), Soft Computing, Multimedia Tools and Applications, and Computational Economics.

Many researchers employ data-processing techniques in conjunction with DL models for enhanced forecasting performance. Time series decomposition is widely used to extract underlying patterns by separating components of varying frequencies (Akşehir & Kılıç, 2024, 2024; Ali et al., 2023; Fan & Zhang, 2024; Guo et al., 2024; Li et al., 2024, 2024c; Qi et al., 2023; Rezaei et al., 2021; Song et al., 2023; Wen et al., 2024; Yan et al., 2020; Yañez et al., 2024). Similarly, dimensionality reduction techniques, such as PCA, help preserve data variation while reducing complexity (Ding et al., 2023; Mansoor et al., 2025; Sarıkoç & Celik, 2024; Wang et al., 2023). Given the sensitivity of DL models to parameter configurations, various optimization methods are employed, including Grid Search (Abdullah et al., 2024; Jin & Zhang, 2024; Kang et al., 2022; Sakib & Mustajab, 2024; Santoro & Grilli, 2022), PSO (Kumar & Haider, 2021; Wang, Yuan, et al., 2024), Adaptive PSO (Kumar et al., 2022), ABC (Kumar et al., 2022), FPA (Kumar & Haider, 2021), ARO (Gülmez, 2023; Salahshour et al., 2024), Bayesian Optimization (Jayanth & Manimaran, 2024; Ping et al., 2024; Yilmaz & Yildiztepe, 2024), GA (Zhang et al., 2023), and Random Search (Touzani & Douzi, 2021).

Over the past five years, while standalone models remain widely used, often serving as benchmarks, the landscape of DL–based financial forecasting has increasingly shifted towards adaptive hybrid architectures that integrate temporal, spatial, and attention mechanisms into unified frameworks.

As illustrated in Fig. 15, standalone models outnumber hybrid approaches, accounting for 61% (215 implementations) compared to 39% (139 implementations) for hybrids. The outer circle highlights the dominance of LSTM-based models in both categories, representing 38% of standalone and 45% of hybrid implementations. Notably, the more specialized CNN-LSTM hybrids comprise 16% of hybrid models. GRU-based architectures are also prevalent among standalone models (12%) but account for approximately 6% within hybrid configurations.

Our findings align with prior review papers (Sezer et al., 2020), which documented the dominance of RNN-based models in financial prediction from 2005 to 2019. Specifically, in our analysis, RNN variants (including Vanilla RNN, LSTM, BiLSTM, GRU, and RC) constitute 56.28% (121 implementations) of standalone models and 51.08% (71 implementations) of hybrid models (primarily LSTM- and GRU-based). They constitute a significant portion of model usage, reinforcing the field's continued reliance on temporal modeling techniques. It is this nuanced interplay among model types, rather than raw counts, that drives ongoing innovation, a theme explored further in Section 8.

Fig. 14 illustrates model adoption trends across forecasting domains. LSTM and BiLSTM dominate across stock (31), index (25), and cryptocurrency (12) forecasting. GRU models are also preferred, while traditional statistical methods (e.g., ARIMA, SMA) and classical ML models (e.g., RF, SVR) remain common benchmarks. Among hybrids, LSTM-based architectures lead, particularly in index (32), and stock (15) forecasting, followed by CNN-LSTM-based hybrids (8 in the same domains). This pattern highlights

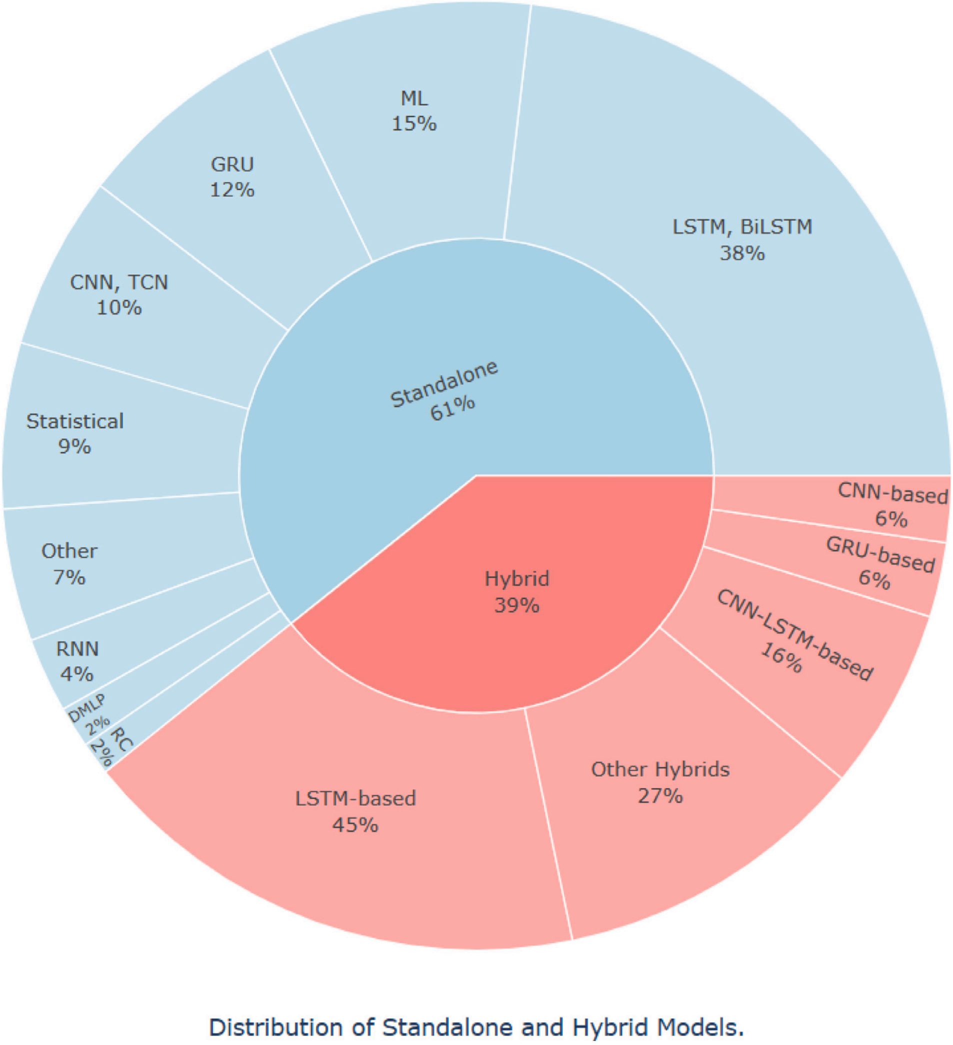

Fig. 15. Nested Double Pie Chart showing model adoption trends. *Note for reviewers: A static image is shown to preserve anonymity. The interactive version is not included, as it is publicly hosted on a personal website and could potentially reveal the authors' identity. The interactive version will be included in the final publication.*

the continued reliance on LSTM-family models for sequential data modeling. GRU-based models provide competitive performance with fewer parameters. The rise of CNN components in hybrid models reflects a growing interest in capturing local patterns and multi-scale features, especially where temporal signals alone may not suffice. Diverse and custom hybrid designs, e.g., GAN-TrellisNet, CEEMDAN-TCN-GRU-CBAM, highlight the field's preference toward customized, task-specific solutions rather than a single, universally superior model.

Notably, RC methodologies remain relatively underexplored, with only four reported implementations mainly in stock and index forecasting. Despite their limited adoption, RC models exhibit strong predictive performance through simple design, computational efficiency, and fast training, underscoring their potential as an effective tool for financial time series forecasting.

Lastly, our analysis of preferred frameworks and development environments, as shown in Fig. 16, underscores an implicit methodological homogeneity: the vast majority of studies employ Python-based stacks. The *Python* category includes tools such as PyCharm Edu, Statsmodels, Tushare, Swarm-PackagePy, yfinance, investpy, seaborn, Matplotlib, Numpy, Pandas, scikit-learn, and PyTorch. The *Other* category includes PostgreSQL15, Google Colab, Apache Hadoop, NVIDIA DIGITS, PyQt5, fbprophet, Math, and Gretl. Across the reviewed studies, Python-based tools dominate the implementations, followed by Keras, and TensorFlow, highlighting their central role in the articles surveyed. However, it is important to acknowledge that some papers do not specify their development environment, which limits the depth of our analysis in this area.

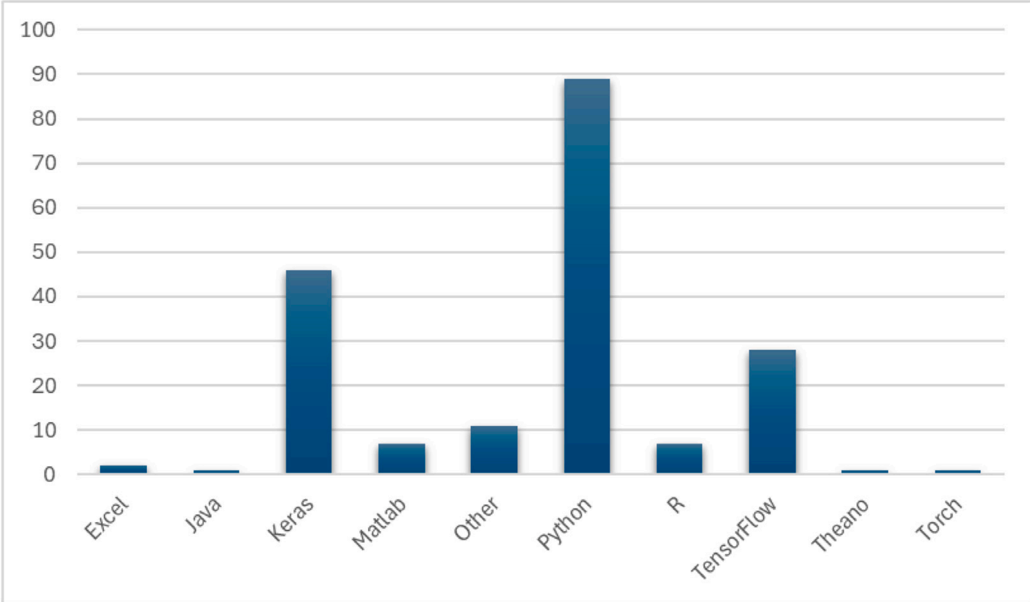

Fig. 16. The preferred development environments. The Python category includes PyCharm Edu, Statsmodels, Tushare, Swarm-PackagePy, yfinance, investpy, seaborn, Matplotlib, Numpy, Pandas, sklearn, and PyTorch libraries. The Other category includes PostgreSQL15, Google Colab, Apache Hadoop, NVIDIA DIGITS, PyQt5, fbprophet, Math, and Gretl.

8. Discussion and open issues

Despite remarkable progress, DL models in financial forecasting still face several limitations in robustness, interpretability, and adaptability to market complexity. As the field matures, a forward-looking research agenda is essential to guide future innovation. In this section, we synthesize key open issues and propose concrete directions for advancing DL research in finance. While certain application-specific opportunities are acknowledged, this review primarily adopts a methodological perspective, emphasizing the design, evaluation, and deployment of DL models in financial forecasting.

8.1. Dominance and underutilization of RNN variants

RNN-based models, especially LSTM and its variants, dominate financial forecasting due to their strength in modeling temporal dependencies. Almost half of the studies adopt LSTM, at least for benchmarking. RC models, being part of the RNN family, remain vastly underexplored despite their computational efficiency and promising predictive performance. This gap suggests a missed opportunity for simpler, faster models in financial applications.

8.2. Feature selection and multimodal data integration

While raw time series data forms the backbone of feature sets, complementary data, such as fundamental indicators (Fischer et al., 2024; Ping et al., 2024), technical factors (Chen et al., 2024; Kalra et al., 2024; Maratkhan et al., 2021; Mohamed Rida et al., 2024; Yang et al., 2020; Zhang et al., 2023), and increasingly, sentiment analysis (Choi et al., 2023; Dhokane & Agarwal, 2024; Fathali et al., 2022; Gangopadhyay & Majumder, 2023; Jung et al., 2024; Kumar et al., 2022; Liu et al., 2024b; Patel et al., 2024; Prachyachuwong & Vateekul, 2021; Song et al., 2020; Srivastava et al., 2023), play vital roles. The rise of financial text mining, leveraging advanced NLP techniques, reflects a growing appreciation for incorporating investor and public market sentiment extracted from news, social media, tweets, forums, and blogs. However, the integration of these heterogeneous data types remains inconsistent and often lacks systematic evaluation of their incremental predictive value.

8.3. Asset focus and emerging domains

Indices and stock prices dominate research focus due to their liquidity and rich historical data, providing stable settings for model evaluation. Meanwhile, cryptocurrency forecasting is emerging rapidly due to its high volatility and lucrative short-term returns. For instance, Bitcoin's price surged from about $12 in late 2012 to nearly $1100 by November 2013, reaching an all-time high above $67,000 in November 2021 (Kraken, 2024). This potential for significant gains drives increased academic and practical interest in cryptocurrencies. Additionally, the continuous, 24/7 nature of cryptocurrency markets, unlike the time-constrained trading hours of traditional stock exchanges, provides richer, more granular data streams, offering greater discriminative and predictive power. This characteristic further reinforces the importance of focused research on these emerging markets (Santoro & Grilli, 2022). Nevertheless, the field must balance enthusiasm for crypto with rigorous validation to avoid hype-driven pitfalls.

8.4. Applications beyond forecasting

DL's application extends into algorithmic trading, portfolio management, and robo-advisory systems. In algorithmic trading, deep reinforcement learning enables more effective buy, sell, and hold strategies by adapting to dynamic market conditions (Ansari et al., 2022; Lee et al., 2021; Tong & Yin, 2022). The integration of financial forecasting with portfolio management, asset allocation, and risk assessment is further evident in robo-advisors, automated financial advisors that design and manage investment portfolios based on a client's financial situation, investment goals, and risk profile (Brenner & Meyll, 2020). Further the increasing availability of high-frequency data, along with advancements in financial sentiment analysis through text mining, enhances decision-making by incorporating investor psychology and market sentiment. Examples of recent advancements in NLP include DistilBERT (Sanh et al., 2019), a faster and more efficient variant of BERT, and BigBird (Zaheer et al., 2020), a sparse attention mechanism for handling long sequences. These developments drive the ongoing automation of trading and investment strategies, positioning this branch of quantitative finance as a key area of future growth.

8.5. Handling market shocks and global crises

Several studies have examined the impact of extreme global events, such as the COVID-19 pandemic, on financial markets (Ampountolas, 2023; Lachaab & Omri, 2023; Nazareth & Reddy, 2024; Widiputra & Juwono, 2024). The heightened market volatility during this period operated as a real-world stress test scenario. This global crisis can be seen as a catalyst for the development of more sophisticated models. The COVID-19 pandemic revealed the limitations of existing models, especially those that relied on historical data and stable market conditions, highlighting the need for more adaptive and resilient approaches that integrate real-time data, financial sentiment, and cross-market signals to better capture and respond to abrupt market shifts.

In response to these challenges, chaos theory-based hybrid models, such as Chaos-LSTM-PR (Durairaj & Mohan, 2021; Durairaj et al., 2023) and Chaos-CNN-PR (Durairaj & Mohan, 2022), show promise by effectively capturing the nonlinear and chaotic dynamics of financial time series. While their structural complexity requires further scrutiny, these models have shown notable improvements in predictive accuracy across diverse asset classes, including index, forex, and commodities forecasting, highlighting the promise of chaos-augmented approaches as a fruitful research direction.

8.6. Quantum computing: A futuristic promise

Quantum computing holds significant theoretical promise to revolutionize DL for financial forecasting by improving the representational power and computational efficiency. It leverages qubits, units of information that can exist in states ranging from zero to one. Unlike classical computing, where units are binary, qubits enable simultaneous exploration of multiple parallel paths in a single calculation unit (Ramezani et al., 2020), through superposition and entanglement. This could substantially improve the speed and accuracy of financial predictions. Quantum algorithms also provide a more comprehensive framework for DL than classical computing, aiding in the optimization of the objective function (Wiebe et al., 2014), which is critical for complex financial modeling and decision-making. However, current quantum hardware falls short due to the limited computational capacity and high error rates (Xu, Wang, et al., 2024). Despite these challenges, the potential for collaborative research at the intersection of DL and quantum computing is immense (Garg & Ramakrishnan, 2020), and active efforts are needed to translate this potential into finance-ready solutions.

8.7. Interpretability and explainability challenges

The black-box nature of DL models remains a significant challenge in terms of interpretability and explainability, hindering users' ability to comprehend and trust the model's predictions. Interpretable and explainable ML techniques arise from the need to design *intelligible* ML systems, i.e. ones that a human mind can comprehend, and to understand and explain the predictions made by opaque models, such as deep neural networks (Marcinkevičs & Vogt, 2020). While there is no universal agreement within the ML community or mathematical rigor on the exact definitions of interpretability and explainability, these are often closely tied and researchers use them interchangeably (Linardatos et al., 2020; Marcinkevičs & Vogt, 2020). However, these are distinct terms. We loosely define interpretability as the capacity to understand a model's internal mechanisms and diagnose its behavior, while explainability requires that models provide explicit reasoning for their decisions and outputs. A more formal definition is given by Gilpin et al. (2018), wherein interpretability refers to the ability to understand what a model did, but on its own it is insufficient for trust in black-box systems. Explainability requires that models offer explicit justifications for their behavior, thereby fostering user trust and clarifying the underlying reasons for their decisions. While explainable models are inherently interpretable, the reverse is not always true. An explanation is assessed by its interpretability, the ability to describe the system's inner workings in a user-friendly language, and its completeness, the ability to describe the operation of a system in an accurate way (Gilpin et al., 2018). Ultimately, the success of interpretability hinges on delivering descriptions that are both simple and meaningful to the user, taking into account the user's cognitive framework, knowledge, and biases (Gilpin et al., 2018). To this end, explainable artificial intelligence (XAI) seeks to enhance the transparency, comprehension, and trustworthiness of complex DL models. Recent efforts have focused in explanation of deep network processing, representation, and system-level explanation generation, yielding promising results (Gilpin et al., 2018). Interpretability and explainability remain active research areas, particularly given the integration of diverse DL models and their increasing deployment in real-world applications such as financial advisory, healthcare, and autonomous vehicles. We remain optimistic that continued interdisciplinary collaboration will facilitate the "opening" of black-box models, enabling deeper understanding, enhanced transparency, and ultimately, greater user trust.

8.8. Lightweight models for real-world deployment

Deploying DL models on smart devices with limited computational resources necessitates lightweight architectures, as traditional models demand substantial processing power, hindering real-time financial predictions on mobile platforms or edge devices. To address this, techniques such as network pruning, model compression, quantization, and knowledge distillation are being utilized to develop efficient models. Network pruning reduces model complexity by eliminating neurons with minimal contributions, while quantization reduces the precision of model weights to save memory. Knowledge distillation involves training a smaller model (student) to replicate the behavior of a larger, more complex model (teacher), thereby maintaining accuracy with fewer resources. Additionally, feature selection helps identify the most relevant input features, further streamlining the model. Collectively, these strategies enable simplified, scalable architectures that maintain high predictive accuracy even when deployed on mobile platforms and edge devices.

8.9. Need for standards in evaluation

A persistent challenge in financial DL research is the lack of standardized evaluation protocols, as emphasized by Rundo et al. (2019), Thakkar and Chaudhari (2021), and Chen et al. (2023). Differences in dataset quality, preprocessing, and validation strategies, such as using clean versus noisy data or applying random splits that inadvertently introduce look-ahead bias, undermine the comparability of results and increase the risk of overfitting. Additionally, publication bias exists in the field, whereby studies reporting positive results are more likely to be published, while negative or null findings are often omitted.

Although benchmark datasets and testing protocols are not fully realistic, often carrying biases such as survivorship or reliance on revised data, they nonetheless are useful for enabling relative comparisons across studies. Standardized benchmarks establish a common testing ground, ensure consistent market histories and train/test procedures, and help distinguish genuine predictive ability from flaws/shortcuts in the experimental design. For future research, we recommend a balanced approach that combines exploratory research on varied datasets and settings, with the use of standardized benchmarks, thereby ensuring both innovation and comparability.

8.10. A design-principles taxonomy

Moving beyond mere model counts and acronyms, we present a taxonomy of financial forecasting architectures based on six design principles:

1. **Sequential-memory frameworks:** Pure recurrent architectures (LSTM, GRU) remain the baseline for smooth, univariate data modeling.
2. **Spatial–temporal hybrids:** CNN–LSTM/GRU and graph-augmented neural networks effectively capture spatial (cross-asset) and temporal (over-time) dependencies, useful for portfolio and market-wide analyses.
3. **Attention-augmented ensembles:** Transformers and attention-enhanced RNNs dynamically adapt to shifting market regimes, albeit at the expense of heavier compute.
4. **Chaos and quantum-inspired hybrid models:** These capture deep nonlinearities and computational parallelism but add complexity.
5. **Preprocessing-augmented pipelines:** Decomposition techniques like EMD, VMD, and wavelets reduce noise and isolate intrinsic modes.
6. **Diverse asset class forecasting:** While stocks and indices remain dominant, crypto forecasting is rising rapidly, offering unique modeling challenges and opportunities.
7. **Multimodal feature integration:** Combining technical, fundamental, and sentiment signals, especially from financial text mining, improves robustness but remains inconsistently applied across studies.
8. **Application-oriented modeling:** DL models increasingly support real-world use cases, including algorithmic trading engines, robo-advisory services, and portfolio optimization tools.

This taxonomy shifts the focus from brand-name architectures toward a design-principles perspective, enabling clearer identification of fertile intersections (e.g., attention-driven, chaos-filtered pipelines) for the next generation of research.

8.11. Critical methodological gaps

Two key shortcomings persist across the literature:

- **Lack of robustness evaluation:** Few models are tested under adverse market conditions or subjected to stress scenarios, raising questions about model resilience in real-world conditions where regime shifts, and extreme events are common. This issue is further compounded by the lack of standardized evaluation protocols, which continues to impede robust model assessment and comparability across studies.
- **Superficial explainability without domain validation:** While XAI methods such as SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017), Local Interpretable Model-agnostic Explanations (LIME) (Ribeiro et al., 2016), and integrated gradients are gaining traction, they are seldom coupled with domain-specific validation practices. Few studies apply XAI primarily to visualize feature importance or to provide superficial interpretability without linking explanations to financial reasoning, expert judgment, or regulatory frameworks. This lack of contextual validation reduces the practical interpretability of the models, especially in high-stakes environments where transparency, accountability, and compliance are essential. Coupling XAI methods with scenario-based stress tests, or regulatory compliance checks could significantly enhance the credibility of AI-driven insights in financial decision-making.

8.12. Initial questions answered

- Which DL models are preferred (and more successful) in different applications?
  Response: While standalone models are more frequently used in applications, hybrid models, when tailored to the characteristics of the dataset, the forecasting task, and the underlying problem, often provide superior performance. Recurrent architectures, particularly LSTM and its variants, remain a dominant and effective choice in financial forecasting, both as standalone models and as core components in hybrid frameworks. CNNs, especially when combined with LSTMs (e.g., CNN-LSTM), are also widely preferred, reflecting their complementary strengths in capturing spatial and temporal dependencies.
- What challenges do DL models face when applied to financial time series data?
  Response:
  - **Data & market challenges:** DL models face significant challenges when applied to financial time series, primarily due to non-stationarity, volatility, chaotic behavior, and noise. These issues are further exacerbated by regime shifts and structural breaks, which hinder model generalization. Data scarcity and quality, especially in emerging or frontier markets, can impair model training and reliability. DL models also suffer from overfitting, especially when applied to high-dimensional, volatile financial datasets without robust validation strategies. These data quality concerns necessitate advanced preprocessing techniques such as decomposition, denoising, and dimensionality reduction.
  - **Model optimization & feature engineering:** Proper hyperparameter tuning and task-specific architectural adjustments are critical. Feature selection (attention mechanisms) is essential for extracting meaningful patterns, while reliance on raw stock data alone often fails to capture the full market picture. Thus, integrating technical, fundamental, and sentiment data is important for richer representation.
  - **Computational considerations:** While complex hybrid models can enhance forecasting accuracy, they also introduce significant computational costs. This can limit real-time applicability and deployment on resource-constrained platforms, highlighting the importance of balancing model complexity with efficiency.
  - **Robustness & extreme conditions:** DL models often lack robustness under extreme market conditions. Their performance can degrade sharply during sudden regime shifts, market shocks, or crisis scenarios, representing a key challenge for reliable out-of-sample forecasting and operational resilience.
  - **Interpretability & explainability:** Despite the increased adoption of explainable AI, many DL models remain largely black boxes, and explainability remains shallow, with limited grounding in financial logic or regulatory context. Addressing this limitation is essential to enhance model transparency and foster user trust.
- What is the future direction for DL research in financial forecasting?
  Response: Addressing the methodological gaps and challenges identified earlier, future DL research in financial forecasting will emphasize developing more adaptive, interpretable, and robust models aligned with real-world financial constraints. Key directions include:
  - **Real-time and continual learning:** Architectures capable of dynamically updating to handle non-stationarity and regime shifts, improving viability in high-frequency, volatile environments.
  - **Robustness and stress testing:** Elevating robustness from an afterthought to a core design principle by incorporating stress-testing protocols simulating extreme market events (crises, flash crashes, geopolitical shocks) to ensure out-of-sample resilience.
  - **Finance-specific explainability:** Enhancing explainability by integrating post-hoc methods like SHAP and LIME within finance-specific validation contexts, such as scenario testing, expert feedback, and regulatory compliance, and developing formal metrics to measure transparency, fidelity under stress, and consistency with financial reasoning for end users such as analysts, auditors, and regulators.

- **Modular hybrid models:** Designing task-specific hybrids balancing predictive accuracy and computational efficiency, particularly for real-time applications like algorithmic trading and crypto markets.
- **Behavioral finance integration:** Incorporating behavioral principles (herding, loss aversion, anchoring) and sentiment-driven signals during volatile or irrational phases can enhance models' predictive power by acknowledging the human element inherent in real-world financial decision-making.
- **Multi-modal data fusion:** Expanding DL applications to underexplored asset classes (e.g., crypto, commodities, emerging markets) while integrating diverse data types (technical, textual, macroeconomic, and alternative data) will improve generalizability and robustness across financial contexts.
- **Adaptive architectures:** Employing meta-learning systems that dynamically switch between submodels based on changing market regimes, and reinforcement learning agents that optimize the balance between prediction accuracy and computational efficiency, will further strengthen forecasting capabilities.
- **Causal inference and transparency:** Embedding causal inference and counterfactual reasoning improves generalization, supports "what-if" scenario planning, and enhances transparency, thereby fostering user trust. Integrating symbolic reasoning with DL, known as neuro-symbolic AI, incorporates domain knowledge and financial constraints, enhancing explainability, supporting decisions aligned with financial principles and regulatory requirements, thereby advancing transparent, reliable, and adaptive financial forecasting.
- **Emerging paradigms:** Exploring Liquid Neural Networks, chaos-inspired, and quantum-enhanced models to enhance performance while managing computational complexity.
- **Lightweight models for real-time deployment:** Lightweight DL models enable fast and efficient real-time applications in algorithmic trading, robo-advisors, and the continuously operating cryptocurrency markets, supporting sophisticated decision-making, automation, and personalization on mobile, edge devices, and low-latency platforms.

In short, the future of DL in finance lies not in ever-deeper architectures, but in designing context-aware, interpretable, and adaptive systems that align with the practical constraints and uncertainties of real-world markets.

9. Concluding remarks

The financial community recognizes the growing potential of DL applications for forecasting, as the volume of research in this area continues to rise at an accelerated pace. In this survey, we review studies published between 2020 and 2024, to offer an overview of the current research landscape. We first describe the most prominent DL models in the field and then categorize the studies based on the implementation areas of financial forecasting. Within each category, we organize the studies based on their focus, distinguishing between standalone and hybrid models, novel approaches, sentiment and technical analysis, and various preprocessing techniques employed, such as dimensionality reduction, decomposition, and optimization.

Despite the long history of financial forecasting, the field continues to evolve with the advent of new DL models, their integration, and the infusion of interdisciplinary methods. This presents ample opportunities for future research. By considering the aforementioned observations, addressing the existing challenges, and introducing innovative techniques, the next generation of DL models can be developed, paving the way for more efficient, intelligent, and automated financial systems.

10. Glossary

ABC Artificial Bee Colony

Acc Accuracy

AE Autoencoder

AEX Amsterdam Exchange Index

AGNN Adversarial Game Neural Network

AM Attention Mechanism

AMH Adaptive Market Hypothesis

AR Autoregressive

ARIMA Autoregressive Integrated Moving Average

ARO Artificial Rabbit Optimization

ARR Average Rate of Return

ATR Average True Range

ATX Austrian Traded Index

AUC Area Under the Curve

BACC Balanced Accuracy

BB Bollinger Bands

BERT Bidirectional Encoder Representations from Transformers

BiLSTM Bidirectional Long Short-Term Memory

BIRCH Balanced Iterative Reducing and Clustering using Hierarchies

BIST100 Borsa Istanbul

BPNN Back Propagation Neural Network

BSE Bombay Stock Exchange

BTC Bitcoin

CAC40 A benchmark French stock market index comprising 40 major companies listed on the Euronext Paris exchange

CART Classification and Regression Tree

CBAM Convolutional Block Attention Module

CBPM Combined Bivariate Performance Measure

CCI Commodity Channel Index

CEEMD Complete Ensemble Empirical Mode Decomposition

CEEMDAN Complete Ensemble Empirical Mode Decomposition with Adaptive Noise

CGAN Conditional GAN

CNN Convolutional Neural Network

CNN-BI CNN with Bar Images

ConvLSTM Convolutional LSTM

CPI Consumer Price Index

CSI China Securities Index

DA Direction Accuracy

DAE Denoising Autoencoder

DBN Deep Belief Network

DE-LSTM Differential Evolution-LSTM

DJ30 Dow Jones 30

DJI Dow Jones Industrial Average

DMLP Deep Multilayer Perceptron

DM Diebold–Mariano test

DQN Deep Q-Network

DRL Deep Reinforcement Learning

DS Directional Symmetry

Dstat Directional Change Statistic

DT Decision Tree

DTR Decision Tree Regression

DWT Discrete Wavelet Transform

EEMD Ensemble Empirical Mode Decomposition

EINS Excitatory and Inhibitory Neuronal Synapse Unit

EMA Exponential Moving Average

EMD Empirical Mode Decomposition

EMH Efficient Markets Hypothesis

EMV Ease of Movement

ES Expected Shortfall

ESN Echo State Network

ES-LSTM Exponential Smoothing-LSTM

ETFs Exchange-Traded Funds

ETH Ethereum

ETS Exponential Smoothing State Space Model

EVS Explained Variance Score

EWMA Exponentially Weighted Moving Average

EWT Empirical Wavelet Transform

FCHI Ticker symbol for CAC40 Index

FI Force Index

FMHP Fully Modified Hodrick–Prescott

FPA Flower Pollination Algorithm

FPR False Positive Rate

FTS Financial Time Series

FTSE Financial Times Stock Exchange

GA Genetic Algorithm

GAF Gramian Angular Field

GAN Generative Adversarial Network

GARCH Generalized Autoregressive Conditional Heteroskedasticity

GDAXI A leading German stock market index consisting of 40 major companies trading on the Frankfurt Stock Exchange

GEM Index 399006.SZ

GJR-GARCH Glosten–Jagannathan–Runkle GARCH

GJR-MEM Glosten–Jagannathan–Runkle Multiplicative Error Models

GM Geometric Mean

GRU Gated Recurrent Unit

GSO Glowworm Swarm Optimization

HMM Hidden Markov Model

HONN Higher-Order Neural Network

HSI Hong Kong Hang Seng Index

HTCSI Halal Tourism Composite Sentiment Index, Country-specific

ICA Independent Component Analysis

ICEEMDAN Improved Complete Ensemble Empirical Mode Decomposition with Adaptive Noise

IDBN Improved Deep Belief Network

IMFs Intrinsic Mode Functions

IPSA The Índice de Precio Selectivo de Acciones, Chile's primary stock market index, includes the 30 stocks with the highest average annual trading volume on the Santiago Stock Exchange

IRRLSTM Improved Recurrent Rider LSTM

IXIC Nasdaq Composite

JKSE Jakarta Stock Exchange Composite Index

KLSE Kuala Lumpur Stock Exchange

KNN K-Nearest Neighbors

KOSDAQ Korea Securities Dealers Automated Quotation

KOSPI200 Korean Composite Stock Price Index

KSE Karachi Stock Exchange

LASSO Least Absolute Shrinkage and Selection Operator

LNN Liquid Neural Network

LR Linear regression

LRC Logistic Regression Classifier

LSTM Long Short-Term Memory

LSTM-BN LSTM with Batch Normalization

LTC Litecoin

LTCs Liquid Time-Constant Networks

MA Moving Average

MACD Moving Average Convergence Divergence

MAE Mean Absolute Error

MAPE Mean Absolute Percentage Error

MCS Model Confidence Set

MEMD Multivariate Empirical Mode Decomposition

MF-ADCCA Multifractal Asymmetric Detrended Cross-Correlation Analysis

MF-DFA Multifractal Detrended Fluctuation Analysis

MFNN Multi-Filters Neural Network

MICEX Moscow Stock Exchange

MLR Multiple Linear Regression

MSE Mean Squared Error

mWDN Multilevel Wavelet Decomposition Network

N100 NASDAQ-100

N225 Nikkei 225 Index

N-BEATS Neural Basis Expansion Analysis

NLP Natural Language Process

NMSE Normalized Mean Squared Error

NR Nonparametric Regression

NSE National Stock Exchange of India

NYR Net Yield Rate

NYSE New York Stock Exchange

OHLCV Open, High, Low, Close, Volume

OHLCVC Open, High, Low, Close, Volume, Change

ORELM Outlier-Robust Extreme Learning Machine

PatchTST Patch Time Series Transformer

PCA Principal Component Analysis

PCC Pearson's Correlation Coefficient

PER Price-Earnings Ratio

PR Polynomial Regression

Prophet Prophet

PSO Particle Swarm Optimization

PSR Phase Space Reconstruction

PSX Pakistan Stock Exchange

PTSR Price To Sales Ratio

R^2 Coefficient of Determination

R-GARCH Realized Generalized Autoregressive Conditional Heteroskedasticity

RC Reservoir Computing

RF Random Forest

RL Reinforcement Learning

RMSE Root Mean Square Error

RN Reynolds Number

RNN Recurrent Neural Network

ROC-AUC Receiver Operating Curve-Area Under Curve

ROC Price Rate of Change

RoR Rate Of Return

ROTA Return On Tangible Assets

RSI Relative Strength Index

RVI Relative Volatility Index

S&P/ASX200 S&P/ASX 200 index (XJO)

SARIMA Seasonal Autoregressive Integrated Moving Average

SCI Shanghai Composite Index

SD Standard Deviation

SDE Standard Deviation of Error

Sen Sensitivity

SGD Stochastic Gradient Descent

SGDRegressor Stochastic Gradient Descent Regressor

SHAP SHapley Additive exPlanations

SHOA Spotted Hyena Optimization Algorithm

SHOAGAI-TSF Spotted Hyena Optimization Algorithm with Generative Artificial Intelligence for Time Series Forecasting

Sklearn sklearn

SMA Simple Moving Average

SMI Stochastic Momentum Index

Spe Specificity

SPX S&P 500

SR Sharpe Ratio

SSA Singular Spectrum Analysis

SSCI Shanghai Stock Exchange Composite Index

SSE Shanghai Stock Exchange

SSEC Shanghai Stock Exchange Composite

SSECI Shanghai Stock Exchange Composite Index

SSMI Swiss Market Index

SVM Support Vector Machine

SVR Support Vector Regression

SZCI Shenzhen Component Stock Exchange Index

SZI Shenzhen Component Index

SZSC Shenzhen Component Index

SZSE SZSE Component Index

Tadawul Saudi Arabian stock market

TAIEX Taiwan Stock Exchange Capitalization Weighted Stock Index

TCN Temporal Convolution Network

Theil's U Theil's Inequality Coefficient

TimeGAN Time-Series Generative Adversarial Network

TPMG Three-Point Moving Gradient

TPR True Positive Rate

TRIX Triple Exponential Average

USCI United States Commodity Index Fund

VAR Vector Autoregressive

VaR Value At Risk

VET VeChain

VMD Variational Mode Decomposition

WHTCSI World Halal Tourism Composite Sentiment Index

WT Wavelet Transform

XGBoost Extreme Gradient Boosting

XGBoost Regressor XGBoost Regressor

XJO S&P/ASX200 Index - Australia's leading stock market index, comprising the top 200 companies listed on the Australian Securities Exchange (ASX)

XRP Ripple

CRediT authorship contribution statement

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data will be made available on request.

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